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High-bandwidth absorption-spectroscopy measurements of temperature, pressure, CO, and H_2O in the annulus of a rotating detonation rocket engine

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Abstract

The development and application of two laser-absorption-spectroscopy diagnostics capable of performing high-bandwidth measurements of (1) temperature, pressure, and CO, and (2) temperature and H_2O in the annular combustion chamber of a rotating detonation rocket engine (RDRE) are presented. The mid-infrared (MIR) diagnostic utilized a quantum-cascade laser (QCL) to measure absorbance spectra of three CO absorption transitions near 2008.5 cm^{-1} at 750 kHz using scanned-wavelength direct absorption. Measurements of gas temperature, pressure, and CO partial pressure were obtained from the QCL diagnostic using a nonlinear fitting routine to fit simulated CO absorbance spectra to measured absorbance spectra. The near-infrared (NIR) diagnostic utilized two tunable diode lasers (TDLs) emitting near 7185.6 and 6806.0 cm^{-1} which were modulated at 35 and 45.5 MHz, respectively, and scanned over the linecenters of their respective H_2O absorption transitions to obtain measurements of WMS-4*f*/2*f* and WMS-2*f*/1*f* signals. WMS-4*f*/2*f* measurements were used to infer the collisional-broadening full-width at half-maximum (FWHM) of the H_2O transitions, and the WMS-2*f*/1*f* signals were then used to infer the gas temperature and H_2O partial pressure at up to 1 MHz. The diagnostics were packaged in single-ended sensor assemblies to enable measurements in the annulus of a methane–oxygen RDRE, and results are presented for a test case with the RDRE operating at an equivalence ratio of 1.15 and a total propellant mass flow rate of 0.6 lb/s.

1 Introduction

Rotating detonation engines (RDEs) achieve pressure-gain combustion in a quasi-continuous mode as detonation waves propagate around an annular combustion chamber consuming a mixture of fuel and oxidizer. Detonation improves the thermal efficiency of a fuel-oxidizer mixture compared to deflagration, and can potentially improve the efficiency of combustion-based power generation and propulsion systems [1–4]. Numerous RDE designs have been developed and studied experimentally using thrust measurements [5, 6], pressure measurements with transducers placed in propellant manifolds and the combustion chamber [7, 8], OH chemiluminescence in optically accessible engines [9, 10], and direct high-speed imaging of detonation wave structures [11–13]. While these measurement techniques provide important information about the spatiotemporal evolution of combustion in RDEs, they do not provide quantitative temperature and species measurements.

Laser-absorption-spectroscopy (LAS) diagnostics are valuable for studying combustion chemistry in harsh environments and have been utilized to obtain quantitative measurements of key combustion species in RDEs. Goldenstein et al. [14] performed some of the first laser-absorption measurements of combustion gases in the exhaust of an RDE using near-infrared (NIR) and mid-infrared (MIR) sensors, each consisting of two tunable diode lasers (TDLs). Wavelength-modulation spectroscopy (WMS) was utilized to obtain measurements of temperature and H_2O at 50 kHz and 10 kHz from the NIR and MIR sensors, respectively. Additionally, the authors inferred how nonuniformities in the line-of-sight (LOS) temperature distribution varied with mass flow rate based on the unique energy levels probed. Rein et al. [15] utilized a MEMS-VCSEL to perform the first

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LAS measurements in the annulus of an RDE. A dual-core fiber was used to direct light into the RDE annulus and collect light reflected off the inner body of the RDE providing measurements of H_2O absorbance spectra near 1350 nm to measure temperature and H_2O at 100 kHz. That work built on that of Caswell et al. [16] who used the same fiber bundle and Fourier-domain mode-locked lasers emitting at similar wavelengths to measure temperature, pressure, H_2O mole fraction, velocity, and enthalpy at 50 kHz in a pulse-detonation combustor. Peng et al. [17] later demonstrated the first MIR LAS measurements in the annulus of an RDE using a single-ended sensor to pitch light from MIR TDLs into the RDE annulus and collect back-scattered light directly on a photodetector. The authors utilized WMS with this sensor to yield measurements of H_2O and temperature at 20 kHz. Cassady et al. [18] further developed this technique to perform measurements of H_2O, CO_2, CO and temperature in the annulus of an RDE at 44 kHz. These measurements demonstrated significant progress in studying RDE combustion chemistry via multi-species measurements, but the diagnostics were all limited to measurement rates $\leq$ 100 kHz which is typically too slow to resolve the intra-cycle dynamics, particularly in rotating detonation rocket engines (RDREs).

Recently, several LAS diagnostics capable of performing measurements at rates $\geq$ 1 MHz in harsh combustion environments have been developed. Mathews and Goldenstein [19] developed a new WMS strategy enabling scanned-WMS measurements at 1 MHz and demonstrated measurements in fireballs of energetic materials. The authors modulated the output of two fiber-coupled TDLs at 10s of MHz to provide enhanced noise rejection and enable measurement rates of 1 MHz using scanned-WMS. Recently, Nair et al. [20] developed a scanned-direct-absorption technique enabling measurements of CO, CO_2, temperature, and pressure at 1–3 MHz at the exit plane of an RDRE. This technique utilized bias-T circuitry to enable high-frequency wavelength scanning of a quantum cascade laser (QCL) and an interband cascade laser (ICL) to measure CO's absorption spectrum near 2008.5 cm^{-1} and CO_2's absorption spectrum near 2385.4 cm^{-1}, respectively. The measurements were used to characterize RDRE operation across a range of flow conditions, equivalence ratios, and injector configurations [21].

This work builds on that of Mathews and Goldenstein [19] and Nair et al. [20], and to the best of our knowledge, presents the first measurements of H_2O and and temperature acquired at 1 MHz and the first measurements of CO, temperature, and pressure acquired in the annulus of an RDRE at rates above 50 kHz. To enable these advancements, measurements were performed in a modified RDRE with an extended combustion chamber featuring window ports, and the two diagnostics were packaged in single-ended sensor assemblies similar to those of Peng et al. [17] to deliver and collect light to and from the RDRE combustion chamber. While the QCL diagnostic presented in this work is similar to that of Nair et al. [20, 21] and the measurements were acquired in the same RDRE, several notable difference in both the overall experimental design and in data processing were utilized. These differences include: (1) the use of single-ended sensor assemblies with mounted windows, enabling measurements within the RDRE combustion chamber as opposed to the exhaust plane, (2) the addition of a near-infrared (NIR) sensor for measuring H_2O, and (3) the use of a database-style model for CO absorbance spectra to perform spectral fitting and enable measurements of temperature, pressure, and CO partial pressure. This manuscript presents the design and fundamental operating principles of both diagnostics, the spectroscopic models and data-processing techniques required to perform the measurements, the optical sensor design, and the results of the measurements acquired in the annulus of an RDRE running on methane and oxygen.

2 QCLAS diagnostic for T, P, P_{CO}

2.1 Fundamentals of direct absorption spectroscopy

Direct-absorption spectroscopy (DAS) involves measuring the intensity of light that is absorbed at a given wavelength to infer gas properties such as temperature, pressure, absorbing species mole fraction or partial pressure, and velocity. The intensity of the incident and transmitted light can be related to the spectral absorbance through Beer's law, given in Eq. (1).

$$\frac{I_t(\nu)}{I_0(\nu)} = exp[-\alpha(\nu)]. \tag{1}$$

Here I_0 and I_t are the intensity of the incident and transmitted light, respectively, α is the path-integrated spectral absorbance, and ν [cm^{-1}] is the optical frequency.

In scanned-DAS, the wavelength and intensity of a tunable laser are scanned across a single, or multiple, absorption transitions to provide a measurement of the transmitted light intensity and the absorption lineshape, and measurements in the non-absorbing regions can be used to infer the incident light intensity. By measuring multiple absorption transitions with different lower state energies (E''), the gas temperature can be measured [22]. Once the temperature is known, the absorbing species partial pressure can be found based on the magnitude of the absorbance. Additionally, if the absorption lineshape is measured and the collisional-broadening coefficients and bath gas composition are known or estimated, the

gas pressure can be inferred from the measured collisional-broadening FWHM, $\Delta\nu_c$.

2.2 Absorption spectroscopy model for CO

The spectral absorbance can be related to the gas properties of the absorbing species using Eq. (2).

$$\alpha(\nu) = \sum_j S_{ij}(T) P_i L \phi_{ij}(\nu, \Delta\nu_d, \Delta\nu_c) \tag{2}$$

Here $S_{ij}(T)$ [cm^{-2} atm^{-1}] is the temperature-dependent linestrength of transition j and absorbing species i, P_i [atm] is the partial pressure of the absorbing species i, L [cm] is the path length through the absorbing gas, and ϕ_{ij} [cm] is the spectral lineshape of transition j and absorbing species i, modeled here as a Voigt profile. The Voigt profile is the convolution of the Gaussian profile and the Lorentzian profile which accounts for Doppler- and collisional-broadening, respectively, assuming they are decoupled, with a Gaussian full-width at half-maximum (FWHM) of $\Delta\nu_d$ and a Lorentzian FWHM of $\Delta\nu_c$.

The Doppler-broadening FWHM is related to the gas temperature and is given by Eq. (3):

$$\Delta\nu_d = \nu_0 \left(7.1623 \times 10^{-7}\right) \sqrt{\frac{T}{M}} \tag{3}$$

where ν_0 [cm^{-1}] is the linecenter frequency, T [K] is the temperature, and M [g/mol] is the molecular weight of the absorbing species. The collisional-broadening FWHM is related to the gas pressure and bath gas composition and is given by Eq. (4). The temperature dependence of the collisional-broadening coefficient was modeled using a power law given by Eq. (5).

$$\Delta\nu_c = 2P \sum_A X_A \gamma_{B-A}(T) \tag{4}$$

$$\gamma_{B-A}(T) = \gamma_{B-A}(T_0) \left(\frac{T_0}{T}\right)^n. \tag{5}$$

Here P [atm] is the gas pressure, X_A is the mole fraction of species A, γ_{B-A} [cm^{-1}/atm] is the collisional-broadening coefficient for a transition of species B being broadened by species A, T_0 is a reference temperature (296 K here), and n is the power-law exponent.

Figure 1 shows a simulation of the absorbance spectrum of CO near 2008.5 cm^{-1} at different pressures which highlights the three transitions which dominate CO's absorbance spectra here. The P(2,20), P(0,31), and P(3,14) transitions of CO all exhibit strong absorbance at the expected RDRE operating conditions, where the notation refers to P(v'',J''). The lower state energy, E'', of these transitions

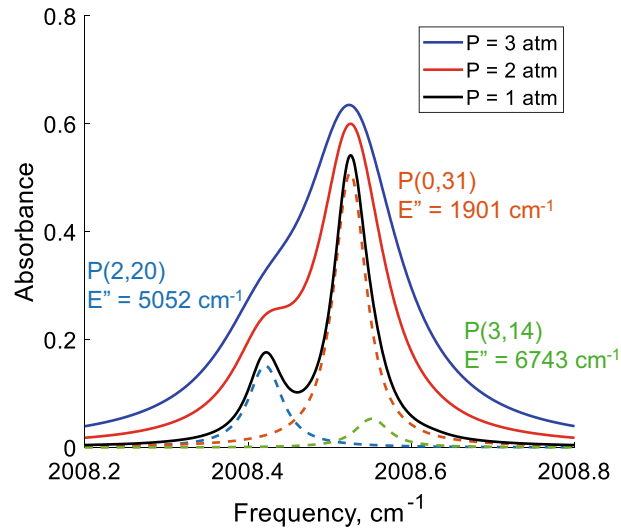

Fig. 1 Simulation of CO absorbance spectrum near 2008.5 cm^{-1} at pressures from 1 to 3 atm. The spectra belonging to the three dominant transitions at these wavelengths are shown individually by dashed lines for the simulation at 1 atm. These simulations correspond to $T = 2500$ K, $X_{CO} = 0.20$, and $L = 1.2$ cm. The bath gas composition was estimated using the procedure described in Sect. 2.2.1, and notably for these conditions $X_{H_2O} = 0.59$.

are 5052, 1901, and 6743 cm^{-1}, respectively. The large difference in E'' between the transitions enables high-sensitivity thermometry to be performed [21, 23]. The P(3,14) transition is blended with the P(0,31) transition at all pressures of interest and the P(2,20) transition becomes blended with the P(0,31) transition at pressures greater than ≈ 2 atm.

Measured CO absorbance spectra were compared to simulated spectra to infer the gas temperature, pressure and CO partial pressure. Rather than fitting Voigt profiles to the individual absorption transitions, the CO absorbance spectrum was simulated using temperature, pressure, and CO partial pressure as free parameters, and the entire spectrum was fit to the measured CO absorbance spectrum. The spectrum was simulated using reference linestrength, linecenter, and lower state energy parameters from the HITEMP database [24] (CO parameters updated in 2019) and following a simulation procedure similar to that of Goldenstein et al. [25]. The absorbance spectrum simulation also included transitions from 2005.0 to 2012.9 cm^{-1} with linestrengths $> 1 \times 10^{-40}$ cm^{-1}/(molecule cm^{-2}) at 296 K to model far-wing absorption of nearby transitions. The linestrength cutoff was employed to reduce the computational cost of the simulation.

The absorbance for each transition was then calculated following Eq. (2) assuming uniform temperature, pressure, and species distributions along the line-of-sight, and the

absorbance from each transition was summed to provide the simulated CO absorbance spectrum. For each transition, the temperature-dependent linestrength was calculated according to Eq. (6):

$$S(T) = S(T_0)\frac{Q(T_0)}{Q(T)}\left(\frac{T_0}{T}\right)\exp\left[-\frac{hcE''}{k}\left(\frac{1}{T}-\frac{1}{T_0}\right)\right] \times\left[1-\exp\left(\frac{-hc\nu_0}{kT}\right)\right]\left[1-\exp\left(\frac{-hc\nu_0}{kT_0}\right)\right]^{-1} \quad (6)$$

where $S(T_0)$ [cm^{-2} atm^{-1}] is the linestrength at the reference temperature T_0 (296 K here), Q is the partition function, h [J·s] is Planck's constant, c [cm/s] is the speed of light, E'' [cm^{-1}] is the lower state energy, and k [J/K] is the Boltzmann constant. The Doppler-broadening FWHM was calculated according to Eq. (3)and the collisional-broadening FWHM was calculated according to Eq. (4). The procedure for calculating the collisional-broadening coefficients and bath-gas composition (for determining the gas pressure) is described in the following section.

2.2.1 Collisional-broadening model

A collisional-broadening model was required to simulate the absorbance spectra in the manner described above and, in general, is required to infer the gas pressure from a measurement of the transition lineshape. The collisional-broadening model defines how $\Delta\nu_c$ is calculated as a function of gas conditions, and it requires knowledge of the bath gas composition and the collisional-broadening coefficient for each transition with each collision partner in the bath gas.

The bath-gas composition was estimated by performing fixed temperature and pressure chemical equilibrium simulations of methane and oxygen at the equivalence ratio prescribed by the RDRE operating condition (1.15 here) with temperatures ranging from 500 to 4000 K and pressures from 0.5 to 10 atm. The resulting concentrations of the combustion products were used to determine the bath-gas composition, albeit with significant uncertainty, as a function of temperature and pressure, which were free parameters in the spectral-fitting routine. Alternatively, the methods of Nair et al. [20] could be used which assumes isentropic expansion of detonation products from the Chapman–Jouguet (CJ) state to 1 atm. While both methods exhibit shortcomings, we chose our method because it allows the bath gas composition to vary based on the temperature and pressure of the mixture. It should be noted that only species with mole fractions > 0.0001 were included in the collisional-broadening model, and for the measured RDRE conditions this included: CO, CO_2, H_2O, OH, H_2, O_2, O, and H. Since the CO partial pressure was also a free parameter of the model, the concentration of CO determined by P_{CO}/P was used instead of the T–P equilibrium value. Additionally, the mole fraction of CO_2 was used to balance the simulated bath gas composition such that the sum of the mole fractions of all included species was equal to one.

Collisional-broadening coefficients were taken from HITEMP [24] or Hartmann et al. [26] when available, else they were calculated following the procedure developed by Nair et al. [20], which is described again here for completeness. CO self-broadening coefficients were taken from the HITEMP database [24] and a power law exponent, n_{CO}, of 0.50 was used to calculate γ at a given temperature using Eq. (5). Collisional-broadening coefficients for CO broadened by CO_2, H_2O, and O_2 were calculated as a function of J'' using broadening coefficients and power law exponents from Hartmann et al. [26] who reported calculated CO collisional-broadening coefficients for a variety of collision partners as a function of J'' at high temperatures. The collisional-broadening coefficients for CO–H_2 broadening were calculated by scaling measured CO–H_2 collisional-broadening parameters for a single transition to adjust for the higher J'' transitions used here. CO–H_2 collisional-broadening parameters for the R(0,11) transition measured by Sur et al. [27] were multiplied by the ratio of CO–N_2 broadening parameters at a given J'' value to those at J''=11 using the CO–N_2 broadening parameters from Hartmann et al. [26]. The collisional-broadening coefficients for CO broadened by OH, O, and H were calculated by scaling the CO–N_2 broadening coefficients from Hartmann et al. [26] by the optical collision diameter of CO with each species Y ($\sigma_{CO\text{-}Y}$) using Eq. (7) [28], and the CO–N_2 power-law exponent was used for these species.

$$\gamma_{CO\text{-}Y(T)} = \gamma_{CO\text{-}N_2(T)}\left(\frac{\sigma_{CO\text{-}Y}}{\sigma_{CO\text{-}N_2}}\right)^2\sqrt{\frac{\mu_{CO\text{-}N_2}}{\mu_{CO\text{-}Y}}}. \quad (7)$$

Here $\sigma_{CO\text{-}Y}$ is the optical collision diameter for CO with species Y, defined by Eq. (8), and $\mu_{CO\text{-}Y}$ is the reduced mass of CO and Y, defined by Eq. (9).

$$\sigma_{CO\text{-}Y} = \frac{1}{2}(\sigma_{CO} + \sigma_Y) \quad (8)$$

$$\mu_{CO\text{-}Y} = \frac{m_{CO}m_y}{m_{CO} + m_y}. \quad (9)$$

Here σ_Y is the optical collision diameter for species Y, and m_Y is the molecular mass of species Y. The optical collision diameters were assumed to be equal to the kinetic collision diameters found in the CHEMKIN transport manual [29]. While this is not accurate, the accuracy of Eq. (7) only requires that the ratio of optical collision diameters for two collision partners is well modeled by the ratio of kinetic collision diameters for the same two collision partners. This is a fair assumption for many collision partners, with H_2 and

H_2O as notable exceptions, which is why these scaling relations were not used for said species.

With the bath gas composition and collisional-broadening parameters defined for the species and transitions of interest, the collisional-broadening model can be used to calculate $\Delta\nu_c$ for any CO transition at an arbitrary T and P, thereby enabling the gas pressure to be determined from the measured spectra. Table 1 lists the collisional-broadening parameters for the three dominant CO transitions with all of the collisional partners that are significant here.

2.3 Data processing

Prior to each test, the laser intensity profile without absorption was recorded to provide a measurement of the incident light intensity, I_0. The transmitted laser intensity, I_t, was recorded during RDRE operation. 15 scans of the measured I_0 were averaged together producing a single reference I_0 measurement that was used to calculate the absorbance spectra for the entire time history of measured I_t. Optical emission from the RDRE was measured during a short period in between each laser scan where the current supplied to the QCL was below its lasing threshold, and the intensity of the optical emission subtracted from the measured I_t for each scan. The absorbance spectrum, $\alpha(\nu)$, for each laser scan was then calculated with peak-normalized measurements of I_t and I_0 using Eq. (1).

Gas properties were found by comparing each measured absorbance spectrum to a simulated absorbance spectrum using a nonlinear fitting routine to find the best-fit spectrum. The fitting routine treated T, P, and P_{CO} as free parameters along with a linecenter shift parameter and two coefficients to perform a linear baseline correction. At each iteration of the fitting routine, free parameters were generated, an absorbance spectrum was simulated according to the model described in Sect. 2.2, and the simulated absorbance spectrum was compared to the measured absorbance spectrum. Iterations were continued following a Levenberg–Marquardt algorithm to vary the free parameters, and the best-fit spectrum yielded the measured values of T, P, and P_{CO}.

Table 1 Parameters used to model the collisional broadening of the dominant CO transitions

Parameter	Transition		
	P(2,20)	P(0,31)	P(3,14)
ν_0, cm^{-1}	2008.42	2008.52	2008.55
γ_{N_2}(296 K), cm^{-1}/atm	0.0526	0.0412	0.0607
n_{N_2}	0.57	0.47	0.65
γ_{CO}(296 K), cm^{-1}/atm	0.0550	0.0430	0.0610
n_{CO}	0.50	0.50	0.50
γ_{CO_2}(296 K), cm^{-1}/atm	0.521	0.0405	0.0636
n_{CO_2}	0.50	0.47	0.53
γ_{H_2O}(296 K), cm^{-1}/atm	0.1199	0.0957	0.1268
n_{H_2O}	0.72	0.61	0.77
γ_{OH}(296 K), cm^{-1}/atm	0.0383	0.0300	0.0442
n_{OH}	0.57	0.47	0.65
γ_{H_2}(296 K), cm^{-1}/atm	0.0610	0.0477	0.0703
n_{H_2}	0.47	0.39	0.53
γ_{O_2}(296 K), cm^{-1}/atm	0.0473	0.0435	0.0500
n_{O_2}	0.56	0.56	0.59
γ_{O}(296 K), cm^{-1}/atm	0.0391	0.0306	0.0451
n_{O}	0.57	0.47	0.65
γ_{H}(296 K), cm^{-1}/atm	0.1001	0.0784	0.1155
n_{H}	0.57	0.47	0.65

3 TDLAS diagnostic for T and P_{H_2O}

3.1 Fundamentals of wavelength-modulation spectroscopy

Wavelength-modulation spectroscopy (WMS) is a laser-absorption-spectroscopy technique where a laser's wavelength and intensity are modulated at a high frequency to encode absorption information at high frequencies located away from noise sources. Lock-in detection is used to extract this absorption information at specific harmonics and reject noise outside of the lock-in filter passband which ultimately leads to improved SNR [28, 30]. Modulating the laser's wavelength and intensity produces harmonic signals which can be used to infer the absorbance at the laser's nominal wavelength and, further, to normalize intensity variations arising from beamsteering, window fouling, or other broadband transmission losses without needing a direct measurement or inference of the incident light intensity. The laser's nominal wavelength can be fixed at a given frequency ("fixed-WMS") to maximize the measurement rate, or scanned over an absorption transition or portion thereof ("scanned-WMS") to improve the measurement accuracy and robustness.

The WMS harmonic signals all scale linearly with laser intensity, and each harmonic is additionally sensitive to different aspects of the measured absorption lineshape. For example, for small absorbance, the WMS-1f signal is primarily sensitive to the measured laser intensity while the WMS-2f signal is primarily sensitive to the absorption magnitude and second derivative of the lineshape. As a result, normalizing the WMS-2f signal by the WMS-1f signal (i.e., WMS-2f/1f) provides a measurement of the absorbance that is insensitive to transmission losses.

Numerous WMS techniques have been utilized to perform measurements in combustion environments due the noise-rejection benefits of WMS and its resilience to nonabsorbing losses [28, 31]. In this work, our previously developed scanned-WMS-2*f*/1*f* sensor employing near-GHz modulation frequencies with TDLs [19] was modified to enable measurements of H_2O absorption at higher pressures with an unknown bath gas composition. Modulating TDLs at near-GHz frequencies enables higher measurement rates to be achieved and enhances the noise-rejection benefits of WMS enabling measurements of small NIR absorbance signals. For example, the SNR of the H_2O measurements reported here is comparable to the CO measurements despite ≈ 20× smaller absorbance for the H_2O spectra. In addition to the high-frequency wavelength modulation, the nominal wavelength of each laser was scanned across the linecenter of their respective absorption transition to measure the WMS-2*f*/1*f* signal at linecenter as a function of time. This technique, termed "peak-picking WMS" [14], utilizes wavelength scanning to ensure the WMS-2*f*/1*f* signal at linecenter is measured as the transition linecenter shifts due to pressure changes or if the laser's nominal wavelength drifts. Peak-picking WMS was used since it provides improved accuracy over fixed-WMS techniques and enables larger measurement rates compared to full-spectrum scanned-WMS [14].

The WMS-2*f*/1*f* signal at the transition linecenter scales linearly with the transition's integrated absorbance for an optically thin gas ($\alpha < 0.05$), and measurements of the WMS-2*f*/1*f* signal at linecenter can, therefore, be used to calculate gas properties. In the optically thin limit, the WMS-2*f*/1*f* signal at linecenter, $S_{2f/1f}(\nu_0)$, is given by Eq. (10):

$$S_{2f/1f}(\nu_o) \approx \frac{1}{i_0\pi}\int_{-\pi}^{\pi} S(T)P_iL \times \phi(\nu_o + a\cos\theta)\cos(2\theta)\mathrm{d}\theta, \quad (10)$$

where i_0 is the DC-normalized intensity modulation amplitude, ν_0 is the transition linecenter frequency, and a is the modulation depth (i.e, modulation amplitude) of the laser. Since the WMS-2*f*/1*f* measurement at linecenter is directly proportional to the absorbance, relationships similar to those used to calculated gas properties in scanned-DAS can be used to calculate properties with a measurement of the WMS-2*f*/1*f* signal at linecenter. For example, the ratio of WMS-2*f*/1*f* signals at linecenter from transitions with different E'' can be used to measure temperature, and the absolute magnitude of the WMS-2*f*/1*f* signal from a single transition can then be used to measure the absorbing species partial pressure. WMS-2*f*/1*f* signals at linecenter can be simulated as a function of gas properties and compared to the measured ratio of WMS-2*f*/1*f* signals and the WMS-2*f*/1*f* signal magnitude to measure temperature and absorbing species partial pressure, respectively [32].

In addition to utilizing the first and second harmonics, higher order harmonic signals have been used to provide unique advantages such as reducing WMS background signals [33–36]. Sun [35] demonstrated that the WMS-4*f*/2*f* signal at linecenter is primarily dependent on the collisional-broadening FWHM, and is largely insensitive to gas properties. This likely results from the WMS-4*f* and WMS-2*f* harmonics having different sensitivities to the absorption lineshape, but similar sensitivities to the integrated absorbance (i.e., magnitude of absorbance). In this case, variations in the WMS-4*f* and WMS-2*f* signals resulting from changes in gas properties such as temperature and species concentration will cancel out when the WMS-4*f*/2*f* signal is calculated. However, variations in the WMS-4*f* and WMS-2*f* signals due to changes in the absorption lineshape (e.g., resulting from collisional broadening) will not cancel out producing changes in the magnitude of the WMS-4*f*/2*f* signal.

Figure 2 shows the WMS-4*f*/2*f* signal at linecenter simulated as a function of $\Delta\nu_c$ at different temperatures from 2000 to 3000 K, pressures from 3 to 5 atm, and H_2O concentrations from 0.1 to 0.2 by mole. The WMS-4*f*/2*f* signal primarily depends on $\Delta\nu_c$, but a slight temperature dependence at small collisional widths is also noted. This temperature dependence can be accounted for by iteratively calculating the temperature for each measurement until the temperature converges. The relationship between the WMS-4*f*/2*f* signal at linecenter and $\Delta\nu_c$ can be utilized to provide an in situ measurement of $\Delta\nu_c$ in an environment with an unknown bath gas composition and pressure. Since the WMS-2*f*/1*f*

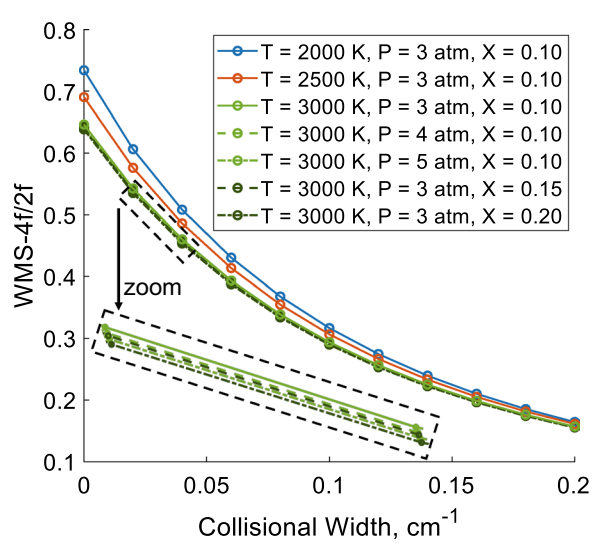

Fig. 2 WMS-4*f*/2*f* signal at linecenter plotted as a function of $\Delta\nu_c$ for temperatures from 2000 to 3000 K, pressures from 3 to 5 atm, and H_2O concentrations from 0.1 to 0.2 by mole

signal is sensitive to the absorption lineshape, a model or measurement of the collisional-broadening FWHM is required to perform a simulation of WMS-2*f*/1*f* signal and accurately measure gas properties. As a result, the WMS-4*f*/2*f* signal was used to measure the instantaneous $\Delta\nu_c$ of a transition to accurately utilize WMS-2*f*/1*f* measurements to infer gas properties.

3.2 Calibration-free WMS model

The absorbance spectrum of two H_2O transitions near 7185.6 and 6806.0 cm^{-1} were measured with two TDLs using a combination of WMS-2*f*/1*f* and WMS-4*f*/2*f*. A spectroscopic model describing the absorbance spectra of H_2O near these transitions was used with a calibration-free WMS model to simulate WMS harmonic signals and infer the gas temperature and H_2O partial pressure. The H_2O transitions near 7185.6 and 6806.0 cm^{-1} have E'' values of 1045 and 3291 cm^{-1}, respectively, and have been used extensively to characterize high-temperature combustion gases because of their strong temperature sensitivity, isolation from interfering absorption transitions, and the availability of spectroscopic parameters that have been validated at high temperatures [14, 19, 37]. The spectroscopic model describing the H_2O absorbance spectrum near these transitions used reference linestrengths provided by Goldenstein et al. [14] for the primary transitions of interest near 7185.6 and 6806.0 cm^{-1}. Linestrength values from the HITRAN2016 database [38] were used to simulate additional transitions from 7184.1 to 7187.0 cm^{-1} and from 6805.2 to 6807.1 cm^{-1} to account for far-wing absorption from neighboring transitions which can become significant at elevated pressures. The collisional-broadening FWHM was treated as an input to the spectroscopic model, and therefore, a sub-model describing the collisional broadening as a function of bath gas composition and pressure was not required. The spectroscopic model was simplified by assuming the collisional FWHM was equal for all transitions from 7184.1 to 7187.0 cm^{-1} and equal for all transitions from 6805.2 to 6807.1 cm^{-1}.

To calculate gas properties using a combination of WMS-4*f*/2*f* and WMS-2*f*/1*f*, the WMS harmonic signals were simulated as a function of the gas temperature, absorbing species partial pressure, and the collisional-broadening FWHM of each transition and compared to measured WMS signals. Several calibration-free WMS models have been developed to perform measurements of gas properties using WMS [32, 39, 40]. The calibration-free model developed by Rieker et al.[32] with an extension reported by Sun [35] was used to simulate the WMS-2*f*/1*f* and WMS-4*f*/2*f* harmonic signals here. In this WMS model, the transmission spectrum of a gas is described as a Fourier expansion with basis functions at the harmonics of the modulation frequency, and the coefficients of each term are used to simulate WMS harmonic signals. The Fourier coefficients are described by Eqs. (11) and (12):

$$H_0(T,P_i,\bar{\nu},a) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \exp\left[-\sum_j S_j(T)P_iL \times \phi_j(\bar{\nu}+a\cos\theta)\right]\mathrm{d}\theta \tag{11}$$

$$H_k(T,P_i,\bar{\nu},a) = \frac{1}{\pi}\int_{-\pi}^{\pi} \exp\left[-\sum_j S_j(T)P_iL \times \phi_j(\bar{\nu}+a\cos\theta)\right]\cos(k\theta)\mathrm{d}\theta, \tag{12}$$

where H_k is the coefficient for the term corresponding to the kth harmonic of the modulation frequency, $\bar{\nu}$ is the nominal laser frequency, a is the laser modulation depth, $S_j(T)$ is the temperature-dependent linestrength of transition j, P_i is the partial pressure of the absorbing gas i, L is the path length through the absorbing gas, and ϕ_j is the lineshape of transition j. The lineshape was modeled as a Voigt profile with the collisional width, $\Delta\nu_c$, treated as an input of the model. The WMS-1*f* and WMS-2*f* harmonic signals are each split into *X*- and *Y*-components which are defined in terms of the Fourier coefficients and given by Eqs. (13)–(16):

$$X_{1f} = \frac{G_{1f}\bar{I}_0}{2}\left[H_1 + i_0\left(H_0+\frac{H_2}{2}\right)\cos\psi_1 + \frac{i_2}{2}(H_1+H_3)\cos\psi_2\right] \tag{13}$$

$$Y_{1f} = -\frac{G_{1f}\bar{I}_0}{2}\left[i_0\left(H_0-\frac{H_2}{2}\right)\sin\psi_1 + \frac{i_2}{2}(H_1-H_3)\sin\psi_2\right] \tag{14}$$

$$X_{2f} = \frac{G_{2f}\bar{I}_0}{2}\left[H_2 + \frac{i_0}{2}(H_1+H_3)\cos\psi_1 + i_2\left(H_0+\frac{H_4}{2}\right)\cos\psi_2\right] \tag{15}$$

$$Y_{2f} = -\frac{G_{2f}\bar{I}_0}{2}\left[\frac{i_0}{2}(H_1-H_3)\sin\psi_1 + i_2\left(H_0-\frac{H_4}{2}\right)\sin\psi_2\right]. \tag{16}$$

Here G_{kf} is the electro-optical gain of the detector at the kth harmonic (1, 2, or 4) of the modulation frequency, $\bar{I}_0$ is the

nominal laser intensity incident on the detector, i_0 and i_2 are the mean-normalized amplitudes of the first- and second-order components of the intensity modulation, and ψ_1 and ψ_2 are the phase shifts between the intensity and wavelength modulation for the first- and second-order components, respectively. Typically when performing WMS-2*f*/1*f* measurements with modulation frequencies < 1 MHz, the gain term ($G_{kf}\bar{I}_0/2$) cancels when normalizing the second harmonic by the first harmonic, and therefore, can be set equal to 1 when simulating harmonic signals. However, the electro-optical gain, G_{1f} and G_{2f}, of a detector must be accounted for when there are frequency-dependent changes in the gain between the first- and second-harmonic components. The frequency-dependent gain of a detector can be significant when the WMS harmonic approaches the bandwidth of the detector, and further discussion is provided in [19].

The magnitude of the WMS harmonic signals were calculated by taking the root-sum-square of the *X*- and *Y*-components of the given harmonic (e.g., WMS-1$f = \sqrt{X_{1f}^2 + Y_{1f}^2}$). The WMS-2*f* signal can be directly divided by the WMS-1*f* signal to obtain the WMS-2*f*/1*f* signal when background subtraction is not required. Nonlinear laser tuning, etalons in the optical system, and ambient absorption outside of the test gas will produce background WMS-2*f*/1*f* signals which can be subtracted from the total WMS-2*f*/1*f* signal using vector subtraction following Eq. (17):

$$\text{WMS-}2f/1f = \left[\left(\left(\frac{X_{2f}}{R_{1f}}\right)_{\text{raw}} - \left(\frac{X_{2f}}{R_{1f}}\right)_{\text{bg}}\right)^2 + \left(\left(\frac{Y_{2f}}{R_{1f}}\right)_{\text{raw}} - \left(\frac{Y_{2f}}{R_{1f}}\right)_{\text{bg}}\right)^2\right]^{1/2}, \tag{17}$$

where R_{1f} corresponds to the root-sum-square of the 1*f* components (i.e., the WMS-1*f* signal), and the subscripts *raw* and *bg* correspond to the raw WMS signal (i.e., total WMS signal due absorption in the test gas and from background sources), and the background WMS signal, respectively. Background WMS signals can be simulated by assuming zero absorption when calculating Fourier coefficients using Eq. (12) (i.e., $P_i = 0$).

Sun developed an extension of this model enabling direct measurements of the collisional-broadening width of a transition using the WMS-4*f*/2*f* signal [35]. The *X*- and *Y*-components of the WMS-4*f* signal can be defined in terms of the Fourier coefficients using Eqs. (18) and (19), respectively, and the WMS-4*f* signal is given by the root-sum-square of the X_{4f} and Y_{4f} components.

$$X_{4f} = \frac{G_{4f}\bar{I}_0}{2}\left[H_4 + \frac{i_0}{2}(H_5 + H_3)\cos\psi_1 + \frac{i_2}{2}\left(H_6 + H_2\right)\cos\psi_2\right] \tag{18}$$

$$Y_{4f} = \frac{G_{4f}\bar{I}_0}{2}\left[\frac{i_0}{2}(H_5 - H_3)\sin\psi_1 + \frac{i_2}{2}\left(H_6 - H_2\right)\sin\psi_2\right]. \tag{19}$$

The spectroscopic and WMS models described above enable the WMS-2*f*/1*f*, WMS-2*f*, and WMS-4*f* harmonic signals to be simulated as a function of T, P_{H2O}, $\Delta\nu_c$, and laser tuning parameters. Simulating these harmonic signals enables the gas temperature and H_2O partial pressure to be inferred from measurements of the WMS-2*f*/1*f* and WMS-4*f*/2*f* signals without a priori knowledge of $\Delta\nu_c$.

3.3 WMS data processing

WMS-2*f*/1*f* and WMS-4*f*/2*f* signals were measured for two transitions near 7185.6 and 6806.0 cm^{-1} using two frequency-multiplexed TDLs (Sect. 4.2). The background laser intensity time history was recorded before each test to provide a measurement of the background WMS-1*f*, WMS-2*f*, and WMS-4*f* signals corresponding to each laser. The laser intensity time histories recorded during RDRE operation and background laser intensity time histories were passed through lock-in filters with bandwidths of 1.1 MHz to extract the WMS harmonic signals according to the procedure presented by Goldenstein et al. [40]. Intensity time histories were multiplied by reference cosine and sine waves at the frequency of interest for a given harmonic and low-pass filtered to provide the *X*- and *Y*-components, respectively, of a given harmonic signal.

Background-subtracted WMS-2*f*/1*f* signals for each laser were calculated using the *X*- and *Y*-components of each harmonic according to Eq. (17). Background-subtracted WMS-4*f* and WMS-2*f* signals were calculated using standard vector subtraction of the *X*- and *Y*-components before calculating the root-sum-square of the *X*-and *Y*-components. The peak WMS-2*f*/1*f*, WMS-2*f*, and WMS-4*f* signals for each scan across the transition linecenter were selected from the WMS signal time histories, and the peak locations for the WMS-2*f*/1*f* signals were used as a reference to select the WMS-4*f* peak corresponding to the transition linecenter (see Fig. 13). The background-subtracted peak WMS-4*f* and WMS-2*f* signals were time-averaged to an effective bandwidth of 100 kHz to improve the SNR of the WMS-4*f*/2*f* signal. The resulting peak WMS-4*f* signal was normalized

by the peak WMS-2*f* signal to provide the measured peak WMS-4*f*/2*f* signal. Measured peak WMS-2*f*/1*f* signals were divided by the ratio of detector gains between the second and first harmonics (i.e., G_{2f}/G_{1f}) to correct for the frequency-dependent gain of the detector. Similarly, measured peak WMS-4*f*/2*f* signals were divided by G_{4f}/G_{2f}.

Peak WMS-2*f*/1*f*, peak WMS-2*f*, and peak WMS-4*f* signals were simulated as a function of T, P_{H2O}, $\Delta\nu_c$ for each set of transitions providing lookup tables used to infer gas properties. Parameters describing the laser's intensity- and frequency-response to high-frequency injection-current modulation (i.e., i_0, i_2, a, ψ_1, ψ_2) were characterized according to the procedures described in [19]. The instantaneous value of i_0 was found as function of time to accurately account for up-scan and down-scan differences in i_0 and the resulting difference in WMS harmonic signals. The measured laser parameters were used with Eqs. (12)–(19) to create a lookup table of background-subtracted WMS-2*f*/1*f*, WMS-2*f*, and WMS-4*f* signals as a function of T, P_{H2O}, and $\Delta\nu_c$. It should be noted that the gain term $(G_{kf}\bar{I}_0/2)$ in all X- and Y-component equations was set equal to 1 when simulating WMS signals, since the frequency-dependent gain of the detector was corrected for by scaling the measured WMS signals during post processing.

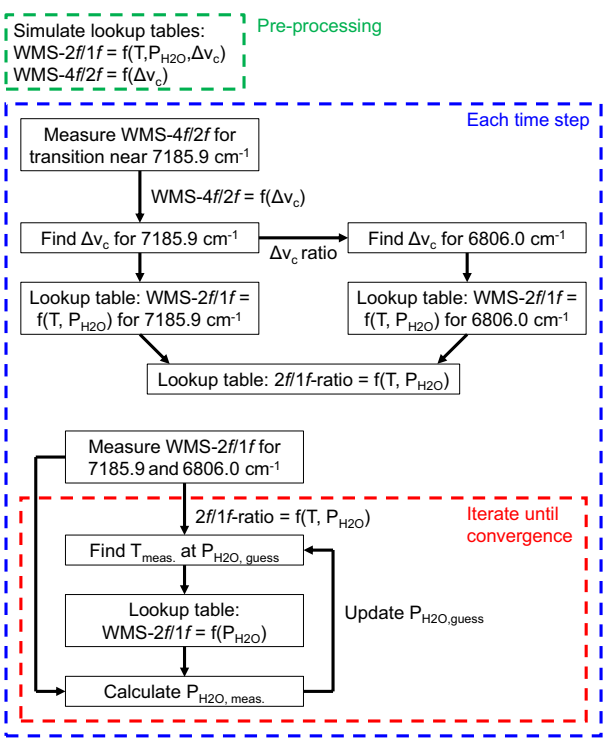

Fig. 3 Flowchart illustrating the iterative data-processing procedure used to calculate the gas temperature and H_2O partial pressure using WMS-2*f*/1*f* and WMS-4*f*/2*f* measurements

Figure 3 shows a flowchart illustrating the data processing technique used to calculate the gas temperature and H_2O partial pressure using WMS-2*f*/1*f* and WMS-4*f*/2*f* signals measured at linecenter for each scan over the absorption transitions. It should be noted that all measured and simulated WMS-2*f*/1*f* and WMS-4*f*/2*f* signals used in calculations were background-subtracted according to the procedures described previously, but for clarity the "background-subtracted" distinction will no longer be explicitly stated. Initial guesses of $T = 2500$ K and $P_{H2O} = 1$ atm were used to initiate the iterative procedure. The measured WMS-4*f*/2*f* signal for the laser near 7185.6 cm^{-1} was compared to the simulated WMS-4*f*/2*f* signals to determine a measured $\Delta\nu_c$ for the respective transition, and the $\Delta\nu_c$ for the transition near 6806.0 cm^{-1} was calculated by multiplying $\Delta\nu_c$ for the transition near 7185.6 cm^{-1} by a factor of 0.676. The value of this scaling factor was determined based on simulations of H_2O and N_2 collisional broadening for these transitions at the expected RDRE conditions. This approach was required to determine $\Delta\nu_c$ for the transition near 6806.0 cm^{-1} due to the low SNR of the WMS-4*f*/2*f* signal of the transition near 6806.03 cm^{-1} as the 4*f* signal (182 MHz) was well past the detector 3 dB bandwidth. The measured values of $\Delta\nu_c$ defined lookup tables of WMS-2*f*/1*f* as a function of only T and P_{H2O}. The ratio of measured WMS-2*f*/1*f* signals between the laser near 6806.0 and 7185.6 cm^{-1} was then compared to the ratio of simulated WMS-2*f*/1*f* signals at the measured $\Delta\nu_c$ and the initial guess value of P_{H2O} to determine a measured temperature. With $\Delta\nu_c$ and T defined, the measured WMS-2*f*/1*f* signal for the laser near 7185.6 cm^{-1} was compared to simulated WMS-2*f*/1*f* signals to find the measured P_{H2O}. The calculation of T followed by P_{H2O} was iterated until the parameters converged within 0.01% between iterations resulting in the measured values of T and P_{H2O}. Typically only two iterations were required.

4 Experimental setup and procedure

4.1 RDRE hardware

The RDRE used for testing was developed by GHKN Engineering, LLC and has been widely studied under a variety of operating conditions, injector designs, and chamber configurations at the Air Force Research Laboratory (AFRL) [11, 41–44]. The combustion chamber annulus had an outer diameter of 76 mm and a 5 mm annular gap. A custom inner body and outer body extension ring were used with the RDRE to extend the length of the combustion chamber by 22.2 mm resulting in a total combustion chamber length of 98.4 mm. The extended inner body included polished flats every 90° and the outer body extension ring featured optical access ports every 90° at corresponding locations to enable

the implementation of laser diagnostics. Methane and oxygen were delivered to the combustion chamber from separate gas manifolds via 72 pairs of impinging injectors at an angle of 30° with respect to the chamber axis, and the injectors were sized to ensure choked flow of methane and oxygen at all operating conditions tested. The RDRE was operated without a nozzle or any flow constriction downstream from the combustion chamber. Each test was initiated by injecting a single detonation wave from H_2/O_2 predetonator tangentially to the combustion chamber 8.9 mm downstream from the injector face after methane and oxygen flows were established [45]. Test times were limited to 500 ms in total duration to ensure that the RDRE hardware and diagnostic instrumentation survived repeated tests. Operation of the RDRE was shutdown by replacing methane and oxygen flows with a nitrogen flow through both gas manifolds.

4.2 Optical setup

Figure 4 shows a schematic of the experimental setup used to perform measurements of temperature, pressure, CO partial pressure, and H_2O partial pressure in the combustion chamber annulus. Single-ended optical sensor assemblies with a design similar to that of Peng et al. [17] were used to direct fiber-coupled laser beams into the combustion chamber annulus and collect light that was back-scattered from the polished flat on the combustion chamber inner body. Figure 5 shows a section view of a single sensor assembly illustrating the beam path and providing a detailed view of the window plugs. Copper window plugs formed the base of each sensor assembly, and face seal O-rings were utilized to

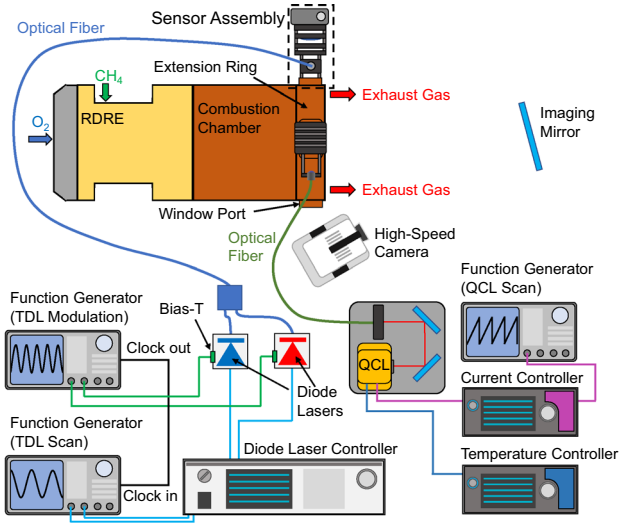

Fig. 4 Schematic of the experimental setup used to acquire LAS measurements of temperature, pressure, P_{CO}, and P_{H_2O} in the RDRE annulus

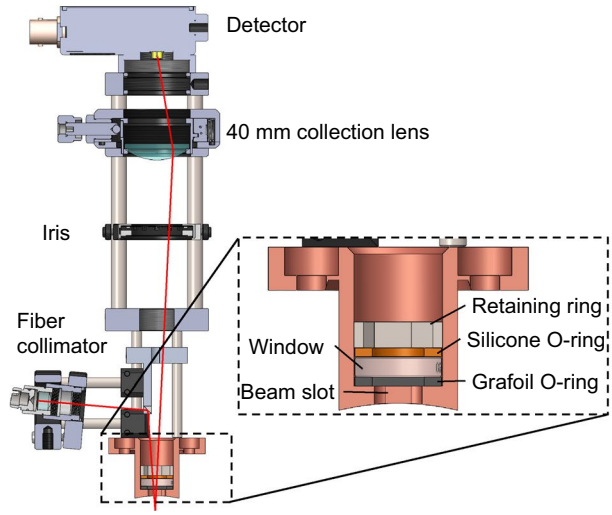

Fig. 5 Single-ended sensor assembly and window plug designs used for LAS measurements in the RDRE annulus

seal the plugs. The inner face of each copper plug sat flush with the inner surface of the RDRE extension ring to minimize flow disruption and featured a 3 × 7 mm machined slot for the beam to pass through. 12.7 mm diameter windows were inserted from the outside of the copper plug, placed between a graphite O-ring (Grafoil) on the combustion chamber side and a silicone O-ring on the outer side, and secured with a threaded retaining ring. This design reduced the frequency of window failure and facilitated fast window replacement.

The MIR diagnostic and the NIR diagnostic were housed in separate sensor assemblies and mounted 90° apart on the extension ring. In each sensor assembly, a fiber-coupled beam was collimated to a beam diameter of ≈ 1.1 mm (Thorlabs F028APC-4950, Thorlabs F110APC-1310), reflected off a 3 mm prism mirror (Thorlabs MRA03-G01), and passed into the combustion chamber through an uncoated, 12.7 mm diameter, wedged sapphire window (Thorlabs WW30530). Light was reflected off the polished flat on the extended inner body, passed through a 6.5 mm diameter iris, collected with a 25.4 mm diameter, 40 mm focal length lens, and directed to either a 250 MHz bandwidth, amplified, MCT detector (VIGO PVMI-2TE-8) for the MIR sensor or a 150 MHz bandwidth, amplified, InGaAs detector (Thorlabs PDA10CF) for the NIR sensor. Signals from each detector were recorded with a 2-channel, 12-bit DAQ (GaGe CSE123G2) with a sample rate of 1.5 GS/s. Careful alignment of the optical sensor assembly was performed to ensure that the beam back-reflected from the uncoated window was spatially filtered by the iris and not incident on the detector, which could result in systematic measurement errors [17]. The body of the NIR sensor assembly was covered

with aluminum tape and the sensor assembly was purged with a constant N_2 flow to reduce absorption and therefore background WMS-2*f*/1*f* signals caused by ambient H_2O.

The MIR diagnostic utilized a QCL (Alpes Lasers) to scan over a portion of CO's absorbance spectrum near 2008.5 cm^{-1} at 750 kHz with a wavelength scanning amplitude of 0.297 cm^{-1}. The laser's temperature and current were controlled with commercial laser controllers (Wavelength Electronics TC5 Lab and QCL500 Lab), which enabled sufficient injection-current tuning with a 750 kHz sawtooth waveform applied through the current controller. While the QCL's wavelength was scanned at 750 kHz, only a portion of the laser's wavelength scan was required to measure the absorbance spectrum, and the diagnostic therefore achieved a 0.6 μs time resolution with a 750 kHz repetition rate (see Fig. 7). The free-space beam emitted from the QCL was coupled into a single-mode, InF_3 fiber using a five-axis fiber port to enable delivery to the RDRE. Before each RDRE test the laser intensity incident on the detector was measured to provide a measurement of I_0 and enable absorbance measurements and the calculation of T, P, and P_{CO} following the procedure described in Sect. 2.3.

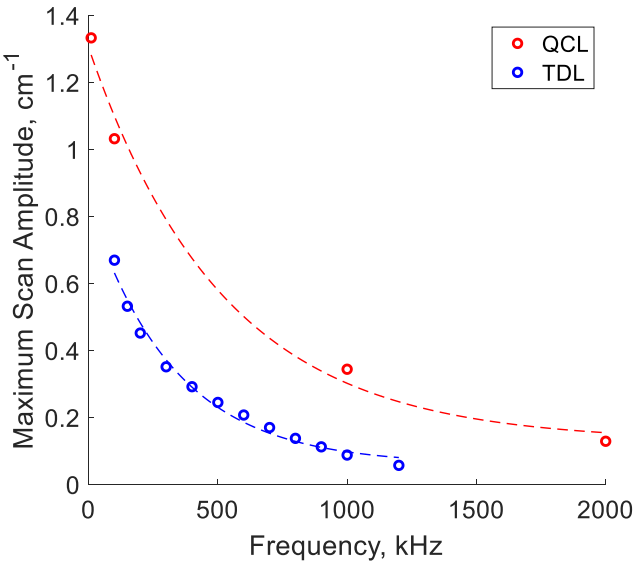

Fig. 6 Maximum scan amplitude of QCL and TDL used to perform measurements in the annulus of the RDRE

Figure 6 shows the maximum scan amplitude that can be achieved using this QCL and applying an injection current through the QCL500 Lab current controller as a function of scan frequency. The maximum scan amplitude of the TDL used for H_2O measurements is shown for reference. At high scan frequencies ($\approx$ 1 MHz), an injection current applied through the QCL500 Lab current controller produced a wavelength scan amplitude comparable to that produced using a bias-T circuit with a similar QCL [20]. Bias-T circuits enable larger scan amplitudes to be achieved but can increase the risk of laser damage, since they bypass maximum current protections within the current controller. It should be noted that the exact wavelength tuning amplitude can vary between lasers due to manufacturing differences even if identical electronics are used to control the laser current and scan the laser's wavelength. Additionally, after performing these measurements, the authors discovered a technique to increase the wavelength scan amplitude of this system by $\approx$ 20% at 1 MHz which has recently enabled

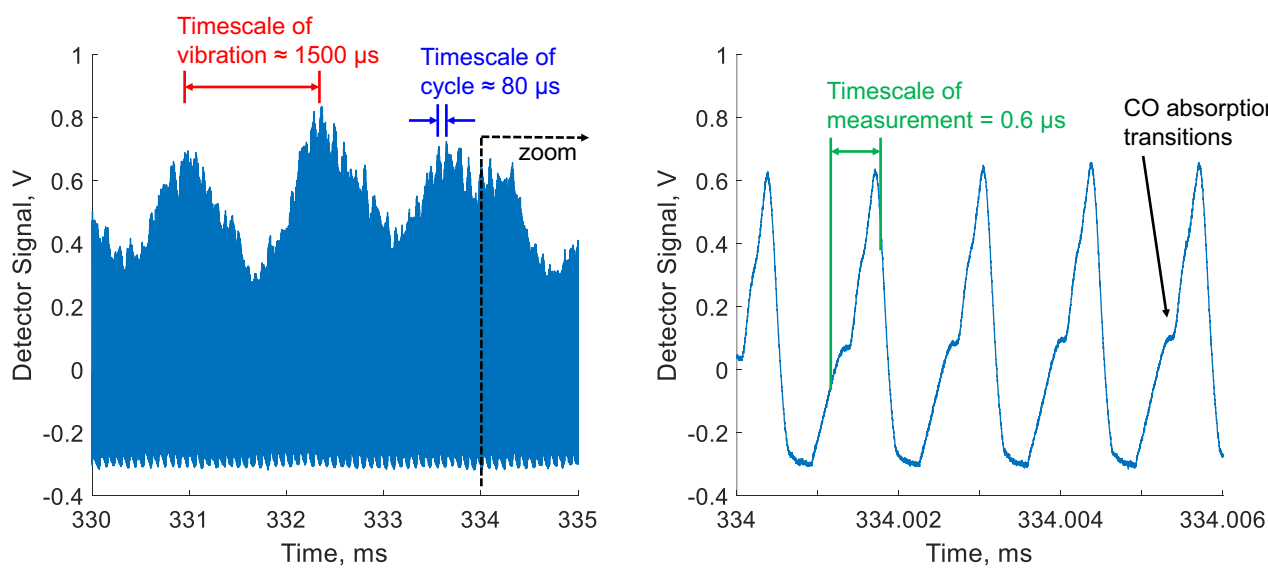

Fig. 7 Time histories of the raw detector signal from QCL diagnostic shown for 5 ms (left) and 6 μs (right). The timescales of mechanical vibrations in the system, RDRE detonation wave cycles, and the QCL measurement are indicated

measurements at 1 MHz in post-detonation fireballs of high explosives in our lab.

The NIR H_2O diagnostic utilized two TDLs (NTT Electronics) to scan over the absorption transition linecenter of two H_2O transitions near 7185.6 and 6806.0 cm^{-1} while the lasers were simultaneously modulated at 35 and 45.5 MHz, respectively, utilizing the frequency-multiplexing configuration described in [19]. The lasers were mounted in butterfly mounts with bias-T inputs (ILX Lightwave LDM-4984-BTB) and the output of the lasers were combined into a single fiber using a fiber combiner/splitter. Injection-current modulation was applied to the lasers by passing 35 and 45.5 MHz sinusoidal waveforms from an arbitrary function generator (Tektronix AFG-3252) through the bias-T inputs on the lasers' mounts providing modulation amplitudes (i.e., modulation depths) of 0.111 and 0.166 cm^{-1} for the TDLs near 7185.6 and 6806.0 cm^{-1}, respectively. The intensity and nominal wavelength of the lasers were simultaneously scanned over the H_2O absorption transition linecenters with a 500 kHz sine wave applied through the current controller using a second arbitrary function generator (AFG). The outputs of the two AFGs were synchronized using a 10 MHz reference clock signal generated with the function generator supplying the modulation signals. Before each RDRE test the laser intensity incident on the detector was measured to provide a measurement of the WMS background signals which were subtracted according to Eq. (17). After testing, T and P_{H2O} were calculated following the procedure described in Sect. 3.3

A 30 × 30 cm square mirror was placed ≈ 5 m downstream of the RDRE exit to enable high-speed imaging of the detonation waves during RDRE operation. A high-speed camera (Phantom v2512) was directed at the RDRE combustion chamber via the square mirror and focused on the injector plane. Imaging data was recorded at 200 kHz with an integration time of 0.6 μs during startup and steady-state operation of the RDRE.

5 Results

5.1 QCL diagnostic results

Figure 7 shows time histories of the raw signal from the detector used for the QCL diagnostic. Signal fluctuations due to mechanical vibration and density gradients in the combustion products due to detonation wave cycles are visible in the raw data. Mechanical vibrations produced intensity variations on a timescale of ≈ 1500 μs, and the detonation wave cycles produced intensity variations on a timescale of ≈ 80 μs for the data shown here. Figure 7(right) shows a 6 μs zoom view of the raw signal which illustrates the measurement timescale of the QCL diagnostic. While the QCL's wavelength was scanned at 750 kHz, only data from a 0.6 μs window in each scan was used to calculate an absorbance spectrum and infer the temperature, pressure, and CO partial pressure. Therefore, the diagnostic has a time resolution of 0.6 μs, and signal fluctuations from mechanical vibration and beamsteering due to detonation wave cycles were frozen on the timescale of the measurement.

Figure 8 shows a measured CO absorbance spectrum and the resulting best-fit spectrum acquired when the RDRE was operating at quasi-steady-state with an equivalence ratio of 1.15 and a total mass flow rate of 0.6 lb/s. The peak-normalized residuals between measured and best-fit absorbance spectrum are dominated by noise in the absorbance spectrum and are typically less than 5% indicating good agreement between the modeled and measured absorbance spectra. The best-fit spectrum corresponds to a measured temperature of 2914 ± 42 K, pressure of 2.51 ± 0.04 atm, and CO partial pressure of 0.496 ± 0.014 atm where the uncertainties represent a 95% confidence interval provided by the fitting routine.

Figure 9 shows time histories of the measured temperature, pressure, CO partial pressure, and associated 95% confidence intervals measured when the RDRE was operating with an equivalence ratio of 1.15 and a total mass flow rate of 0.6 lb/s. Figure 9(left) shows 4 ms time histories of the measured quantities demonstrating the repeatability of the detonation cycles and indicating that the RDRE is operating in quasi-steady-state. The minimum and maximum temperatures are approximately 2500 and 3200 K, the minimum and maximum pressures are approximately 1.5 and 3.5 atm,

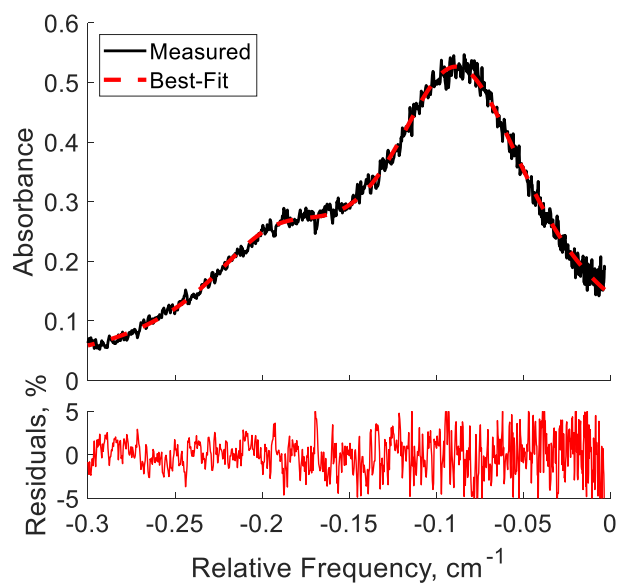

Fig. 8 Single-scan measurement and best-fit absorbance spectra of CO acquired using the MIR diagnostic in the RDRE annulus. The corresponding best-fit gas conditions are T = 2914 K, P = 2.51 atm, and P_{CO} = 0.496 atm

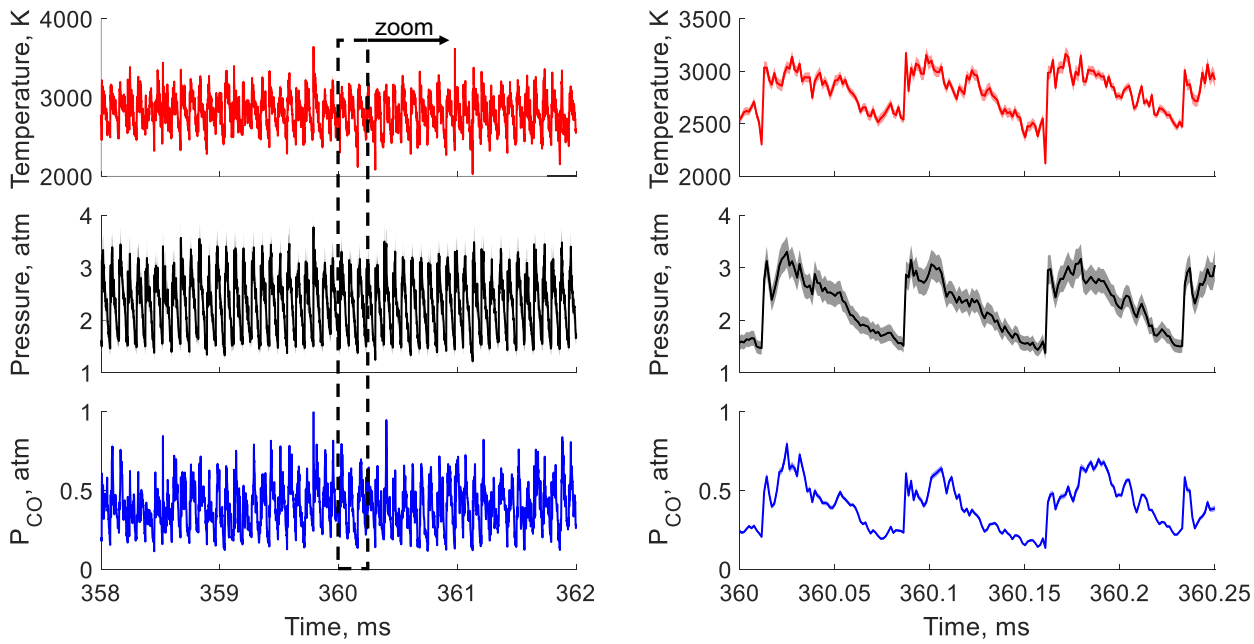

Fig. 9 Time histories of temperature, pressure, and CO partial pressure measured at 750 kHz using the QCL diagnostic. The measurements correspond to the RDRE operating at quasi-steady state (360 ms after ignition) with an equivalence ratio of 1.15 and mass flow rate of 0.6 lb/s

and the minimum and maximum CO partial pressures are approximately 0.2 and 0.7 atm. Figure 9(right) shows 250 μs time histories of the measured quantities illustrating the intra-cycle dynamics resolved in the temperature, pressure, and CO partial pressure measurements. The shaded region in each plot shows the 95% confidence interval in the measured parameter from the nonlinear fitting routine in the temperature and P_{CO} plots. The shaded region shown in the pressure measurement includes the 95% confidence interval from fitting and the additional uncertainty due to the collisional-broadening model. Each detonation wave cycle begins with a sharp increase in temperature, pressure and CO partial pressure as the combustion products produced immediately behind the detonation wave pass the measurement line of sight. Temperature, pressure, and CO partial pressure all decrease in the flow downstream of the detonation wave before the next wave arrives. Frequently, a second peak in the measured T, P, and P_{CO} is observed approximately 15 μs after the initial rise at the arrival of the detonation wave. The timing and amplitude of this 'double-peak' feature in the signal indicates that the second peak is a shock wave, reflected from the exit plane of the combustor. Intercycle variation of the detonation strength and local flow properties at the measurement location leads the observed variation in the T, P, and P_{CO} fluctuation levels.

Figure 10 shows time histories of temperature and CO partial pressure measured near 333 ms (left) and 360 ms (right) and estimations of the gas conditions at the measurement location determined from chemical equilibrium simulations of a methane and oxygen mixture at an equivalence ratio of 1.15. The gas conditions were estimated using two methods: (1) the CH_4/O_2 mixture was expanded isentropically from the CJ state to the measured pressure and equilibrated with constant entropy and pressure to calculate the temperature and CO partial pressure and (2) fixed temperature and pressure (T–P) equilibrium simulations of the CH_4/O_2 mixture were performed at the measured temperature and pressure to calculate the CO partial pressure. The isentropic expansion calculations provide an estimate of the upper boundary for the gas temperature and an estimate of the expected CO partial pressure if all heat release occurred immediately behind the detonation wave and no heat was lost to the combustion chamber walls between the detonation wave and the measurement plane. The fixed T–P equilibrium calculations provide an estimate of the CO partial pressure if the mixture is at local thermochemical equilibrium without a constraint on mixture entropy.

In Fig. 10(left), near 333 ms, the measured temperature generally remains below the temperature predicted assuming isentropic expansion throughout the entire detonation wave cycle. Immediately following the arrival of the combustion products produced directly behind the detonation wave (i.e., at the start of the detonation wave cycle), the measured CO partial pressure agrees with that predicted by the semi-empirical, T–P equilibrium calculations. This indicates that the mixture within the region immediately following the detonation wave is at thermochemical equilibrium. In the combustion products

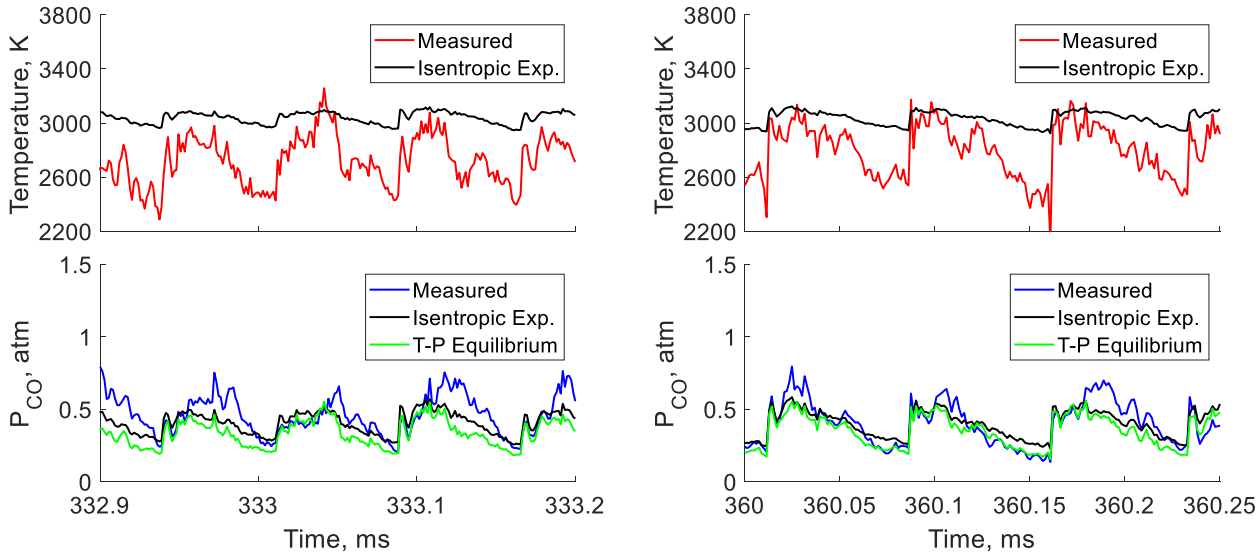

Fig. 10 Temperature and CO partial pressure measured near 333 ms (left) and 360 ms (right) compared with values estimated from isentropic expansion and equilibrium simulations at the measured temperature and pressure

produced further behind the detonation wave, the measured CO partial pressure is significantly larger than that predicted by the semi-empirical equilibrium calculations, indicating incomplete combustion of the mixture or a more fuel-rich reaction zone due to incomplete mixing. The CO partial pressure predicted by isentropic expansion of the detonated CH_4/O_2 mixture is generally closer to the measured CO partial pressure compared to that predicted by fixed T–P equilibrium. However, the measured CO partial pressure still only agrees with the CO partial pressure predicted assuming isentropic expansion early in the detonation wave cycle. These results suggest that the RDRE may not be operating in its most efficient configuration (e.g., injector arrangement, equivalence ratio, mass flow rate) or with the most efficient detonation wave structure at times near 333 ms.

In Fig. 10 (right), near 360 ms, the temperature measured in the combustion products produced immediately behind detonation wave consistently reaches the temperature predicted by isentropic expansion (≈ 3100 K) before falling to values significantly lower than those predicted by isentropic expansion late in the detonation wave cycle. The CO partial pressure measured near 360 ms remains in agreement with the values predicted by both isentropic expansion and fixed T–P equilibrium calculations for a larger portion of the detonation wave cycle compared to the CO partial pressure measured near 333 ms. The measured CO partial pressure agrees with the value predicted by fixed T–P equilibrium early in the detonation wave cycle, shows a small deviation above the equilibrium value at its peak, and quickly returns to the value predicted by equilibrium late in the detonation wave cycle. These results suggest that at times near 360 ms, the RDRE may be operating more efficiently compared to times near 333 ms, which could be the result of changes in the detonation wave structure between these measurement times as discussed further.

5.2 Detonation wave instability

Figure 11 shows spectrograms of the temperature and pressure measured by the QCL diagnostic illustrating the frequency content of each measurement as a function of time. The labels A, B, and C indicate the times corresponding to the three plots shown in Fig. 12. From 320 to ≈ 345 ms, there is strong frequency content at 13.3 kHz corresponding to the fundamental frequency of detonation waves within the RDRE annulus. Second- and third-harmonics of this frequency also appear in both measurements of temperature and pressure. Approximately 345 ms into the test, an additional lower frequency component near 10.4 kHz appears and the frequency of the initial detonation wave structure begins to steadily decrease. This defined change in the frequency content of the temperature and pressure can be correlated with a change in the detonation wave structure in the RDRE annulus. When these frequencies converge at 353 ms, a new detonation wave structure with a frequency of 13.6 kHz emerges. The intermittent presence of a frequency component near 9.0 kHz in the pressure measurements from 320 to 345 ms indicates that detonation wave structure may not

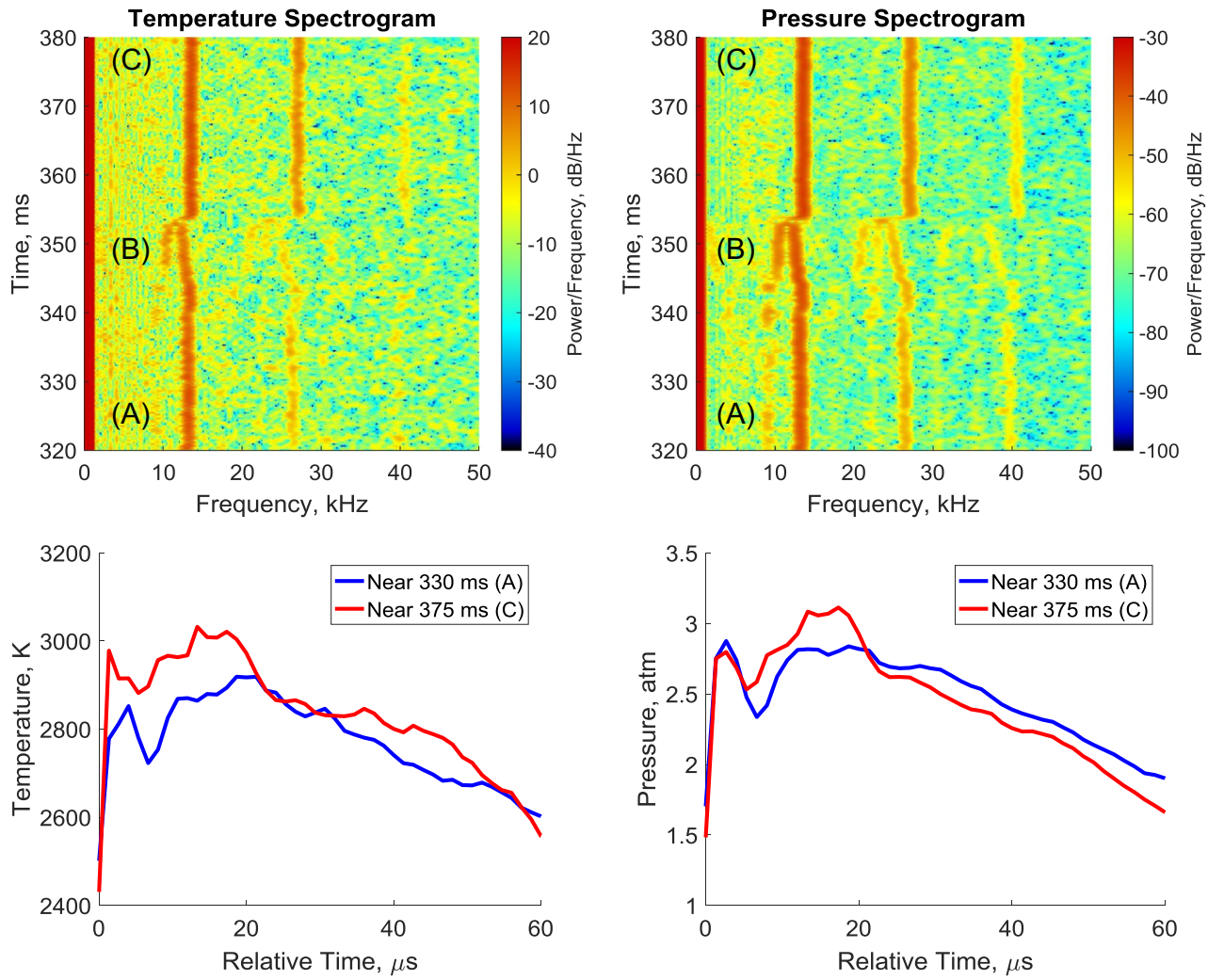

Fig. 11 Spectrograms showing the frequency content of the temperature and pressure measured with the QCL diagnostic as a function of time (top) and plots of the cycle-averaged temperature and pressure near 330 and 375 ms (bottom)

be stable during those times. This instability is confirmed in the region from 345 to 353 ms consisting of multiple frequency components indicating significant shifting in the unstable detonation wave structure. In the final detonation wave structure established at 353 ms, the pressure measurement no longer shows prominent frequency content near 9.0 kHz and the frequency component has a larger peak amplitude of ≈ −37 dB/Hz compared to ≈ −40 dB/Hz from 320 to 345 ms suggesting a more stable detonation wave structure. Figure 11(bottom) shows cycle-averaged temperature and pressure plots near 330 and 375 ms averaged over 60 detonation wave cycles (≈ 5 ms). These plots indicate the the temperature and pressure typically reaches higher values after the new detonation wave structure is established near 353 ms. This supports the results shown in Figure 10 which suggest that the RDRE is operating more efficiently at 360 ms compared to 333 ms. These results also suggest that an absorption sensor can be used to identify instability in the RDRE annulus and potentially be used to predict changes in the detonation wave structure.

Figure 12 shows three theta-t contour plots produced from high-speed imaging data acquired simultaneously with the QCL diagnostic measurements presented in Fig. 11. Theta-t contour plots indicate the angular position of detonation waves in the RDRE annulus as a function of time, and the plots reveal changes in the detonation wave structure corresponding to frequency content in the temperature and pressure measurements. Near 325 ms, Fig. 12(top), there are two detonation waves propagating counterclockwise relative to the imaging system. The dual detonation wave structure is relatively stable and present from 320 ms until ≈ 345 ms. Near 349 ms, Fig. 12(center), there are two sets of counter-propagating detonation waves. These counter propagating sets of detonation waves are not stable within the RDRE

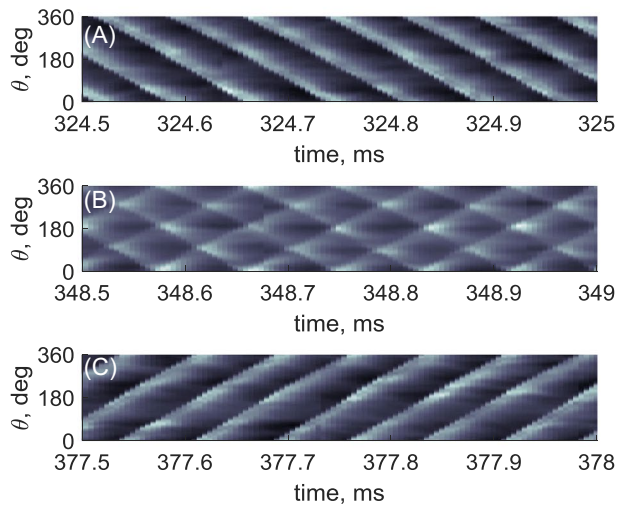

Fig. 12 Contour plots showing the angular position of detonation wave fronts as a function of time

annulus, and the complex structure collapses to a steady two wave configuration at ≈ 353 ms. Near 378 ms, Fig. 12(bottom), there are only two detonation waves propagating clockwise relative to the imaging system illustrating a steady propagating wave structure. These imaging results confirm the shift in detonation wave structure indicated by changes in the frequency content of the measured temperature and pressure (Fig. 11), and they illustrate how high-speed imaging can compliment and improve interpretation of absorption measurements in the RDRE annulus. Additionally, the imaging data confirms that an absorption sensor measuring at a single location in the RDRE annulus can be used to identify stability in the RDRE detonation wave structure.

5.3 TDL diagnostic results

Figure 13 shows time histories of measured WMS harmonic signals acquired in a test case when the RDRE was operating with an equivalence ratio of 1.13 and a total mass flow rate of 0.57 lb/s. Figure 13(left) illustrates the structure of the measured WMS signals where the WMS-2*f*/1*f* signals exhibit a peak each time the laser's nominal wavelength was scanned over the transition linecenter, and the WMS-4*f* signal exhibits multiple peaks in the given wavelength scan envelope. Peak values of the WMS-2*f*/1*f* signals illustrate the up-scan and down-scan differences as the laser's tuning parameters, primarily i_0, change throughout the wavelength scan. The variation between up-scan and down-scan measurements was accounted for during data processing using different laser parameters for up-scans and down-scans when simulating WMS signals. The transition near 7185.6 cm^{-1} shows peak up-scan WMS-2*f*/1*f* signals near 0.023 corresponding to an absorbance of 0.0186, and the transition near 6806.0 cm^{-1} shows peak up-scan WMS-2*f*/1*f* signals near 0.013 corresponding to an absorbance of 0.0102. Peak values of the WMS-4*f* signal were chosen using the location of the selected WMS-2*f*/1*f* peaks for the transition near 7185.6 cm^{-1} as a reference to ensure the WMS-4*f* peak corresponding to the transition linecenter was selected.

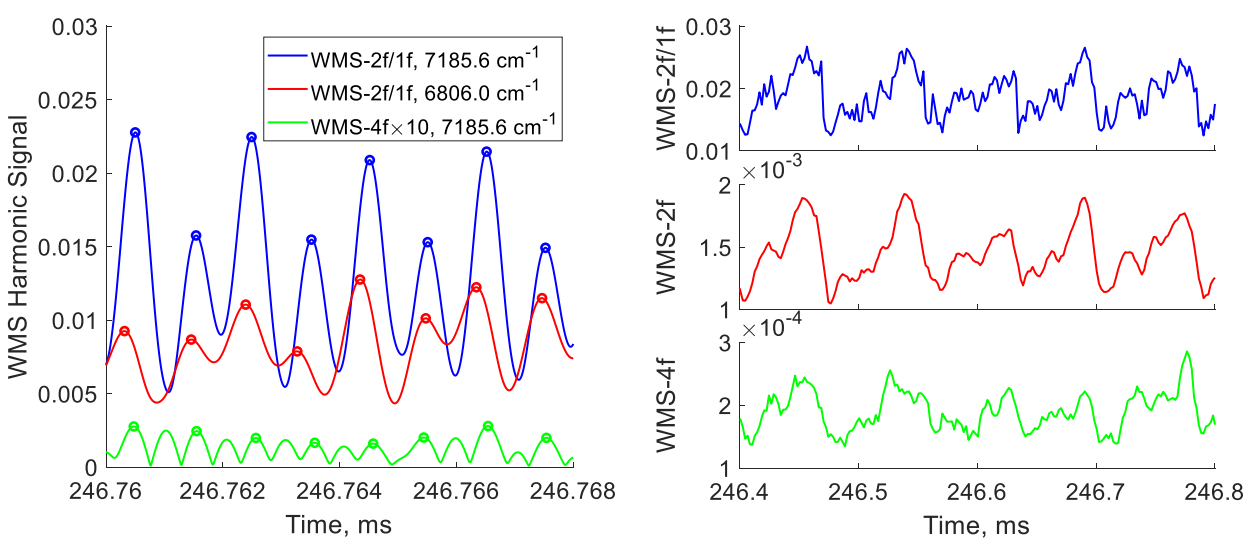

Fig. 13 WMS-2*f*/1*f* signals for transitions near 7185.6 cm^{-1} and 6806.0 cm^{-1} and WMS-4*f* signal for transition near 7185.6 cm^{-1} (left). The peak WMS signals are used for calculation of gas properties. Peak WMS-2*f*/1*f*, WMS-2*f*, and WMS-4*f* signals for the transition near 7185.6 cm^{-1} (right). Only up-scan values are shown for clarity, and the WMS-2*f* and WMS-4*f* signals are filtered to an effective bandwidth of 100 kHz

Figure 13(right) shows peak WMS-2*f*/1*f*, WMS-2*f*, and WMS-4*f* signals for the up-scans over the transition near 7185.6 cm^{-1}. The peak WMS-2*f* and WMS-4*f* signals were used to calculate the collisional-broadening FWHM and the peak WMS-2*f*/1*f* signal was used to calculate the gas temperature and H_2O partial pressure. All three harmonic signals generally increase throughout the detonation wave cycle. This is likely due to decreasing temperature downstream from the detonation wave resulting in an increased fraction of molecules in the absorbing state of the transition near 7185.6 cm^{-1}. For only the $\Delta\nu_c$ measurement, the WMS-2*f* and WMS-4*f* signals were time-averaged to an effective bandwidth of 100 kHz to improve the precision of the collisional width measurement.

Figure 14 shows temperature and H_2O partial pressure time histories measured at RDRE test conditions of $\phi = 1.13$ and a total mass flow rate of 0.57 lbs/s. Time-averaging was applied to the temperature and H_2O partial pressure time histories to improve the measurement precision resulting in a 500 kHz measurement rate. The 1 MHz measurements without time averaging are shown in lighter colors behind the 500 kHz measurements. The temperature measured with H_2O shows the same trends as those seen in the CO results. Specifically, the temperature increases sharply at the arrival of combustion products produced immediately behind the detonation wave and steadily decreases as the gas downstream from the detonation wave reaches the measurement line of sight. The measured temperature consistently reaches peak values of ≈ 3000 K at the beginning of each detonation wave cycle which agrees with the peak temperature measured by the QCL diagnostic. The minimum temperature values of ≈ 1700 K are slightly lower than those measured with the QCL diagnostic which could be the result of measuring H_2O transitions with lower E'' values compared to those of the CO transitions, if temperature nonuniformities are pronounced. Alternatively, this may result from imperfect background subtraction or run-to-run differences. The H_2O partial pressure rises to peak values of ≈ 1.0 atm when the detonation wave arrives and falls to minimum values of ≈ 0.3 atm as the detonation wave passes. Frequency analysis of the measured temperature and H_2O partial pressure time histories provide an RDRE operating frequency of 12.9 kHz which is in reasonable agreement with the frequency of 13.3 kHz measured with the QCL diagnostic for the same nominal operating condition but different test. These results demonstrate the ability of near-GHz WMS to provide measurements of small absorption signals, $\alpha < 0.01$, in detonation environments.

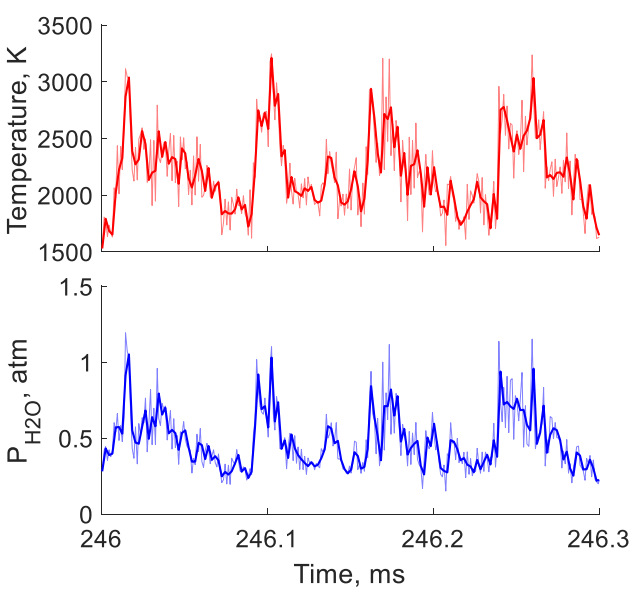

Fig. 14 Measured time histories of temperature and H_2O partial pressure acquired at 1 MHz using the TDL diagnostic. Measurements acquired at 1 MHz are shown in lighter colors, and time averaged measurements with a rate of 500 kHz are shown in darker colors

6 Conclusions

The design and application of two diagnostics capable of performing absorption measurements in the annulus of an RDRE were presented. A MIR diagnostic measured the CO absorbance spectrum near 2008.5 cm^{-1} with a QCL at 750 kHz using scanned-wavelength direct-absorption. The gas temperature, pressure, and CO partial pressure were inferred from the best-fit spectra simulated with a spectral model utilizing HITEMP linestrength parameters and a custom collisional-broadening model. A NIR diagnostic measured two H_2O absorption transitions near 7185.6 cm^{-1} and 6806.0 cm^{-1} using two frequency multiplexed TDLs employing near-GHz scanned-WMS. WMS-4*f*/2*f* harmonic signals were utilized to infer the collisional broadening of the H_2O transitions *in situ*, and WMS-2*f*/1*f* harmonic signals were used to infer the gas temperature and H_2O partial pressure. High-speed images of the detonation waves within the RDRE annulus were recorded to aid in data interpretation. Both diagnostics were capable of resolving intra-cycle dynamics in the combustion chemistry as products produced by repeated detonation waves passed the measurement line of sight. Measurements of CO partial pressure indicated equilibrium of the mixture immediately following the detonation wave but reduced combustion efficiency in the region downstream from the detonation wave. Measurements of temperature and pressure were used to identify instabilities in the detonation wave structure within the RDRE annulus based on the frequency content of the measured gas properties. This work illustrates the potential of these diagnostics for studying a wide variety of highly transient combustion

and propulsion applications where temperature and species measurements with μs time resolution are needed.

Appendix: uncertainty analysis

This paper presented measurements of temperature, pressure, and CO partial pressure obtained with a scanned-DAS diagnostic and measurements of temperature and H_2O partial pressure obtained with a scanned-WMS diagnostic. This appendix presents estimates of the uncertainty in each of these measurements including contributions from model parameters and experimental noise. The Taylor series method of uncertainty propagation assuming independent variables was used for this analysis, and the uncertainty in a given measurement is given by Eq. (20):

$$(\Delta s)^2 = \left(\frac{\partial s}{\partial x_1}\Delta x_1\right)^2 + \left(\frac{\partial s}{\partial x_2}\Delta x_2\right)^2 + \cdots \tag{20}$$

Here, Δs is the uncertainty in the measured parameter s, and x_i are variables with uncertainties Δx_i. The partial derivative sensitivity terms were calculated numerically using the associated scanned-DAS or scanned-WMS model. Uncertainties in each model parameter were taken from the source of the model parameter or estimated using techniques described in the following sections. Additionally, terms quantifying the uncertainty due to noise in the raw data were added in quadrature with the terms quantifying the uncertainty from model parameters.

QCL diagnostic

The QCL diagnostic was used to acquire measurements of temperature, pressure, and CO partial pressure by fitting a simulated absorbance spectrum to a measured absorbance spectrum. Using this technique, uncertainty in the measured values arises from uncertainty in the model parameters used to simulate an absorbance spectrum and from noise in the measured spectrum which has many sources including electronic noise and noise induced from optical losses such as beamsteering and window fouling. Each measured quantity (e.g., T, P_{CO}) is sensitive to different aspects of the measured spectrum, and therefore, each quantity is sensitive to different model parameters as well as noise in the measured absorbance spectra.

Figure 15 shows estimates of the uncertainty in temperature, pressure, and CO partial pressure measured with the QCL diagnostic. Additionally, Fig. 15 shows the sources of uncertainty in each measured parameter and their contributions to the total measurement uncertainty. Reported uncertainty values are the average uncertainty over a series

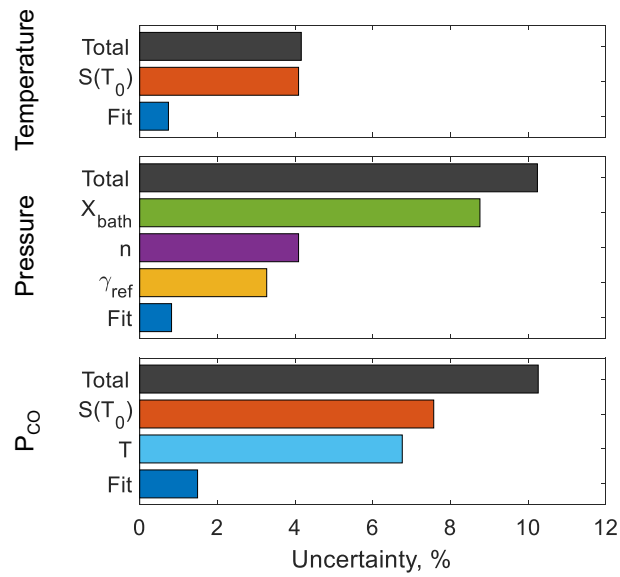

Fig. 15 Uncertainties for temperature, pressure, and CO partial pressure measurements acquired with the QCL diagnostic

of measurements and are shown as a percentage of the given quantity. The following subsections describe how the uncertainty for each parameter was calculated and the sources of uncertainty in model parameters.

Temperature uncertainty

Temperature sensitivity results from measuring spectra associated with transitions with different E'', since the population in states with different E'' has a unique temperature dependence. The ratio of integrated absorbance between two transitions is only a function of temperature which enables temperature to be calculated independently from the absorbing species concentration and gas pressure. While there is some coupling between collisional-broadening and the temperature measurement when performing spectral fitting, the contribution from model parameters other than the reference linestrengths is < 1% and are not shown in this analysis.

The uncertainty in the temperature measurement was found by summing the uncertainties associated with the reference linestrengths of each transition and the uncertainty induced from noise in the data following Eq. (21).

$$\Delta T^2 = \sum_j (\Delta T_{S(T_0)_j})^2 + (\Delta T_{\text{fit}})^2. \tag{21}$$

Here, ΔT is the total measurement uncertainty in temperature, $\Delta T_{S(T_0)_j}$ is the uncertainty in the measured temperature resulting from the uncertainty in the reference linestrength, $S(T_0)$, of transition j, and ΔT_{fit} is the uncertainty in the measured temperature resulting from noise in the measured

absorbance spectrum. ΔT_{fit} is returned from the nonlinear fitting routine and is derived from the sensitivity of the best-fit spectrum to the temperature parameter in the model.

Only the uncertainty from the reference linestrengths for the P(2,20), P(0,31), and P(3,14) transitions of CO were included in this analysis, since the absorbance spectrum is dominated by these transitions at the wavelengths of interest. The uncertainty for the reference linestrengths are tabulated in the HITEMP database [24] and are 5%, 2%, and 10% for the P(2,20), P(0,31), P(3,14) transitions, respectively. The average uncertainty in temperature across a series of measurements resulting from uncertainty in reference linestrengths is 4.1%, the average uncertainty resulting from the model fit is 0.7%, and the average total uncertainty in temperature is 4.2%.

The dominant source of uncertainty in the temperature measurement is the uncertainty in the reference linestrengths, and the uncertainty due to noise in the data is relatively small. To reduce uncertainty in the temperature measurement, precise measurements of the linestrengths of the P(2,20), P(0,31), and P(3,14) transitions of CO must be performed.

Pressure uncertainty

Pressure measurements rely on the measurements of the collisional-broadening FWHM, $\Delta\nu_c$, and the collisional-broadening model describing $\Delta\nu_c$ as a function of gas properties. $\Delta\nu_c$ is not measured directly when using a database style model to fit the measured absorbance spectrum, but the pressure measurements are still derived from $\Delta\nu_c$ of the measured transitions and the collisional-broadening model. Therefore, the uncertainty in the measured pressure results from uncertainty in the collision-broadening model parameters and uncertainty induced by noise in the measured absorbance spectrum. The uncertainty in pressure resulting from noise in the measured absorbance spectrum is returned directly from the nonlinear fitting model and was added in quadrature with the uncertainty associated with each model parameter, similar to Eq. (21).

The collisional-broadening model uses three parameters for each species included in the model: reference collisional-broadening coefficient, γ_{ref}, collisional-broadening power-law temperature exponent, n, and mole fraction of the collision partner, X. Uncertainty in γ_{ref} and n results from variations with the lower state rotational energy and from the scaling analysis used to estimate γ_{ref} for the species without collisional-broadening data reported in the literature [20]. An estimated uncertainty of 5% was used for all broadening coefficients and power-law exponents which is consistent with collisional-broadening uncertainties in the HITEMP database [24]. The uncertainty in the mole fraction of each species was conservatively estimated as the difference between the mole fraction of the given species at the C-J state and the mole fraction defined by T–P equilibrium simulations, as described in Sect. 2.2.1. The average uncertainty in pressure resulting from uncertainty in the model fit, γ_{ref}, n, and X are 0.8%, 3.3%, 4.1%, and 8.8%, respectively, and the average total uncertainty in pressure is 10.2%. The dominant source of uncertainty in the pressure measurement is uncertainty in the concentration of each species used in the collisional-broadening model. Noise in the measured data contributes a relatively small amount of uncertainty to the pressure measurement compared to the collisional-broadening model.

Figure 16 shows the uncertainty in the pressure measurement resulting from uncertainty in the mole fraction, reference broadening coefficient, and power-law exponent of each species used in the collisional-broadening model. The estimated mole fraction of each species is commonly the largest factor contributing to uncertainty in the pressure measurement, and the mole fraction of H_2O has the largest effect. This indicates that the uncertainty in the pressure measurement using the QCL diagnostic could be significantly reduced if H_2O was measured simultaneously along the same line-of-sight.

Despite many similarities between this diagnostic and that of Nair et al. [20], there are differences in the reported total uncertainties in the pressure measurements acquired with these diagnostics. This is due to variations in the methodology used to estimate the uncertainty of model parameters. The approach used here provides a more conservative estimate of the uncertainty in various model parameters, and therefore, produces a larger total uncertainty in the pressure measurement.

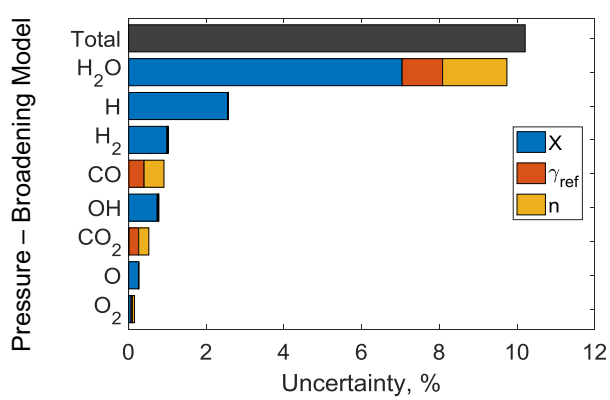

Fig. 16 Uncertainties in the pressure measurement resulting from uncertainty in the mole fraction, reference collisional-broadening coefficient, and power-law exponent of each species used in the collisional-broadening model

CO partial pressure uncertainty

The partial pressure of CO is a function of the magnitude of a transition's integrated absorbance and therefore the overall magnitude of the absorbance spectrum. The CO partial pressure also depends on the measured temperature, since the integrated absorbance is also a function of the linestrength of a transition. While the CO partial pressure measurement exhibits minor coupling with the collisional broadening when a database style model is used for spectral fitting, the collisional-broadening model contributes < 1% uncertainty to the CO partial pressure and is not included in this analysis. The total uncertainty in CO partial pressure includes contributions from the reference linestrengths of the P(2,20), P(0,31), and P(3,14) transitions, the measured temperature, and the noise in the measured absorbance spectrum.

Uncertainty in reference linestrengths were taken from the HITEMP database [24] as described previously. The total uncertainty in the temperature measurement (i.e., 4.2% of the measured temperature) was used to find the CO partial pressure uncertainty resulting from the temperature measurement. The uncertainty in CO partial pressure associated with noise in the measured spectrum was returned directly from the nonlinear fitting routine and was added in quadrature with the other uncertainty terms, similar to Eq. (21). The average uncertainties from model fitting, the temperature measurement, and reference linestrengths were 1.5%, 6.8%, and 7.6% of the measured CO partial pressure, respectively. The average total uncertainty in CO partial pressure was 10.3% of the measured value.

The dominant sources of uncertainty in the CO partial pressure measurement is uncertainty in the reference linestrengths of the CO transitions and uncertainty in the temperature measurement. Noise in the measured spectrum contributes a relatively small uncertainty to the CO partial pressure measurement. Similar to the temperature measurement, more precise measurements of reference linestrengths of CO's P(2,20), P(0,31), and P(3,14) transitions are needed to significantly reduce the uncertainty in the CO partial pressure measurement. This would directly reduce the uncertainty contributed from the reference linestrengths and also reduce the uncertainty contributed from the temperature measurement. The relative uncertainty in CO partial pressure is larger than that of the CO column number density reported by Nair et al. [20]. The differences in relative uncertainties is the result of more conservative approximations used in the uncertainty analysis here, similar to the analysis used for estimating the uncertainty in the pressure measurement.

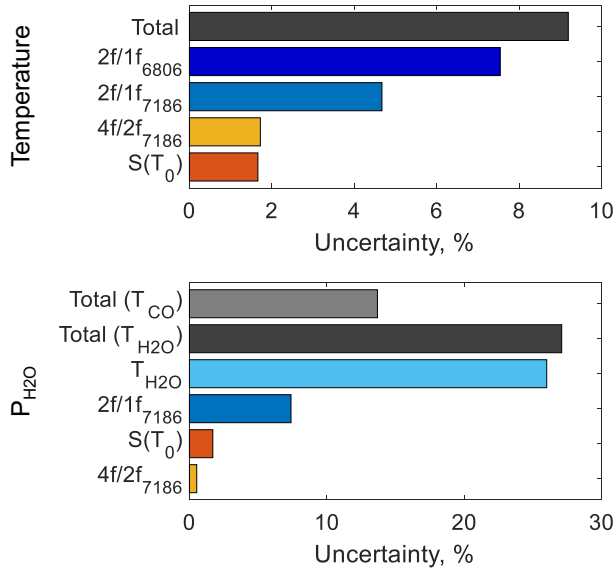

Fig. 17 Average total uncertainty in temperature and H_2O partial pressure measured using the TDL diagnostic. The uncertainty in the H_2O partial pressure measurement performed using the temperature measurement from the QCL diagnostic is shown with the label "Total (T_{CO})"

TDL diagnostic

The TDL diagnostic utilizes calibration-free WMS models and values of WMS-2*f*/1*f* and WMS-4*f*/2*f* signals measured at the transition linecenter to determine the gas temperature and H_2O partial pressure. Using this technique, uncertainty in the measured temperature and H_2O partial pressure results from uncertainty in the model parameters and the noise in the measured WMS-2*f*/1*f* and WMS-4*f*/2*f* signals.

Figure 17 shows estimates of the total uncertainty in temperature and H_2O partial pressure measurements acquired with the TDL diagnostic and the uncertainty in the measured parameter resulting from each source of uncertainty. For all uncertainty estimates presented, the uncertainty was calculated for a series of measurements and averaged to a single value. Additionally, the uncertainty in up-scan and down-scan measurements were calculated independently and averaged. Typically, the uncertainty in up-scan measurements is less than the uncertainty in down-scan measurements, since up-scan WMS-2*f*/1*f* peaks have a larger magnitude compared to down-scan peaks.

Noise in the WMS signals is analogous to noise in the measured absorbance spectrum for the QCL diagnostic and is primarily due to electronic noise. The noise in a given WMS signal was estimated by taking a 5-point moving average of the WMS signal and subtracting the

moving-average from the measured WMS signal to produce a "noise signal". The standard deviation of the noise signal was used as an estimate of the uncertainty for the given WMS signal.

The only spectroscopic parameter required for the calibration-free WMS model used here is the reference linestrength of each transition. A collisional-broadening model and the associated spectroscopic parameters would be required for a peak-picking WMS measurement if only the WMS-2*f*/1*f* signal was measured. However, the instantaneous collisional-broadening FWHM was measured here directly using the WMS-4*f*/2*f* signal. The uncertainty in each measured quantity associated with error in the WMS-4*f*/2*f* measurement is included in this analysis.

Temperature uncertainty

The temperature was measured using the ratio of WMS-2*f*/1*f* signals at linecenter of the transitions near 6806.0 and 7185.6 cm^{-1}. This ratio is primarily only a function of the gas temperature, and any dependence on H_2O partial pressure is accounted for by iterating the calculation of temperature and H_2O partial pressure. The total uncertainty in the temperature measurement includes contributions from the uncertainty in reference linestrength of each transition, the noise in the WMS-2*f*/1*f* signal of each transition, and the noise in the WMS-4*f*/2*f* signal for the transition near 7185.6 cm^{-1}.

The uncertainty in the reference linestrengths for the two transitions of interest near 7185.6 and 6806.0 cm^{-1} were included in this analysis. All other transitions were only modeled to include far-wing absorption, and contribute <1% to the uncertainty in the measured temperature. The linestrengths of the transitions near 7185.6 and 6806.0 cm^{-1} were measured directly by Goldenstein et al. [37, 46] with experimental uncertainties of 2% and <3%, respectively. The uncertainty of each WMS signal was calculated using the moving-average noise measurement described previously, and the total uncertainty in temperature was calculated following Eq. (20) using numerical partial derivatives to find the sensitivity term for each parameter. The relative uncertainty in temperature resulting from uncertainty in reference linestrengths, WMS-4*f*/2*f* signal for the transition near 7185.6 cm^{-1}, WMS-2*f*/1*f* signal for the transition near 7185.6 cm^{-1}, and WMS-2*f*/1*f* signal for the transition near 6806.0 cm^{-1} were 1.6%, 1.7%, 4.7%, and 7.5%, respectively, and the average total uncertainty in temperature was 9.2%.

The dominant source of uncertainty in the temperature measurement is noise in the WMS-2*f*/1*f* signal for the transition near 6806.0 cm^{-1}. This is primarily due to small absorption signals (i.e., $\alpha < 0.01$) for this transition at the gas conditions and path length through the RDRE annulus. Utilizing the WMS-4*f*/2*f* to infer the collisional-broadening FWHM resulted in <2% uncertainty in the temperature measurement which illustrates the value of this technique. Using a collisional-broadening model to estimate the collisional FWHM likely would produce much larger uncertainties in the TDL temperature measurement based on the assumptions required to estimate the bath gas composition and collisional-broadening coefficients.

H_2O partial pressure uncertainty

The H_2O partial pressure was calculated using the magnitude of the WMS-2*f*/1*f* signal for the transition near 7185.6 cm^{-1} after the temperature was found using the TDL diagnostic. The H_2O partial pressure at a given temperature is a function of the magnitude of the WMS-2*f*/1*f* signal, analogous to the integrated absorbance of the transition. Therefore, the measured H_2O partial pressure depends on the measured temperature, the reference linestrength of the transition near 7185.6 cm^{-1}, and the noise in both the WMS-2*f*/1*f* and WMS-4*f*/2*f* signals for the transition near 7185.6 cm^{-1}.

The total uncertainty in the temperature measured with the TDL diagnostic (i.e., 9.2% of the measured temperature) was used to find the uncertainty in the H_2O partial pressure resulting from uncertainty in the temperature. Uncertainty in the reference linestrength for the transition near 7185.6 cm^{-1} was taken from Goldenstein et al. [37] as described previously. The noise in the WMS signals for the transition near 7185.6 cm^{-1} was calculated using the moving-average noise measurement described previously. The total uncertainty in H_2O partial pressure was calculated following Eq. (20) using numerical partial derivatives to find the sensitivity term for each parameter. The relative uncertainty in H_2O partial pressure resulting from uncertainty in the WMS-4*f*/2*f* signal, reference linestrength, WMS-2*f*/1*f* signal, and temperature were 0.5%, 1.7%, 7.4%, and 26.0%, respectively, and the average total uncertainty in H_2O partial pressure was 27.1%.

The dominant source of uncertainty in the H_2O partial pressure measurement is uncertainty in temperature, and this is largely due to low SNR for the weaker transition near 6806.0 cm^{-1}. To reduce uncertainty in the H_2O partial pressure measurement, the gas temperature must be measured with less uncertainty compared to that of the TDL diagnostic in this RDRE. For example, if the H_2O partial pressure was calculated using the temperature measured with the QCL diagnostic, the total uncertainty in H_2O partial pressure would reduce to 13.7%. The total uncertainty in H_2O partial pressure if the temperature from the CO diagnostic was used is shown in Fig. 17 with the label "Total (T_{CO})".

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